Context selection as a partial order: A Blackwell framework and verified LLM evidence

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Abstract

Prompt context is usually chosen by scoring each source with one number and keeping the top few. We study the limits of task-independent scalar rankings for context sources. Two sources can be *incomparable* in Blackwell's sense; we prove that when they are, no single per-source score ranks them correctly for every task. That concerns the ideal user of a source; what a fixed model does with one is separate, and we measure it. Running each source through a battery of pass/fail tasks and comparing pass rates with exact confidence intervals, on six models from four vendors we verify at $\geq 90\%$ in both directions that two hand-built sources cross in realized pass rate. On a designed trap a lexical score, two dense retrievers and a cross-encoder prefer the wrong source, while a listwise LLM reranker reading the deciding detail does not. A strict superset of a sufficient source also lowers pass rates, verifiably on all twelve cells and as far as 100% to 0%; a character-matched padding arm and a reversed-order arm leave that collapse unexplained. Results hold for the evaluated snapshots, prompts and decoders on a hand-built battery; scaling the implied rule to large corpora remains open.

Keywords: context selection; Blackwell order; comparison of experiments; incomparability; partial order; retrieval-augmented generation; large language models; distribution-free verification; anti-monotonicity; prompt engineering

1 Introduction

Suppose a coding assistant is handed one of two short notes about a software module. The first note says how to *call* its main function: the order of the arguments, a keyword that is mandatory, and the error it raises on failure. The second note says how the module *encodes* numbers: a digit format ending in a checksum. For a task that calls the function the first note is decisive and the second is useless; for a task that encodes a number it is the reverse; and neither note can be reconstructed from the other. The two notes are *incomparable* in the sense of Blackwell's comparison of experiments [1], and the consequence is immediate: no single relevance score ranks them correctly for every task, because a score must crown one winner and here neither wins outright. In the body these are the sources W_1 and W_2.

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information, $\mathcal{V}$-usable information [2], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. This paper asks what a task-independent scalar ranking can and cannot represent, and answers it with a classical apparatus from statistical decision theory.

What is and is not new. The mathematics is classical: the garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam's deficiency. So are several of the phenomena, each in its own literature: that utility rather than topical relevance is the right target [3] and is model-dependent [4]; that adding sources can lower accuracy while coverage saturates [5]; that rerankers are steered by lexical similarity [6]; that keeping a non-dominated set is already an operator over heuristic scores [7]; and that the Blackwell and Le Cam apparatus was brought to inference-time context selection concurrently with this work, over which we claim no priority [8]. Section 2 treats each in turn. The gap is that none of them gives the order itself an object or a test: cross-task non-transfer is described as a nuisance rather than read as a partial order, and no finite-sample procedure decides on a live model whether two particular sources are incomparable. That is what C1–C3 below supply.

One gap needs care. Blackwell's theorem is about the *best possible* use of a source, but every number we measure comes from one fixed model: $\mathrm{PASS}_M(W,T) = \mathbb{E}[v_T(\delta_M(T,w))]$, the pass rate of a single rule δ_M that is not optimal. So we state the order twice, for the ideal user and for the fixed model, and prove a one-way bridging lemma whose hypotheses are assumed, not measured; our sources' structural incomparability rests instead on their construction (see Section 4); and we call our estimator a *value-deficiency surrogate*, not a Le Cam deficiency (see Section 6).

Contributions. The paper makes one claim, that the order over context sources is *partial* rather than total, and contributes the three pieces that claim needs: what follows from it, the instrument that makes it testable, and the evidence that it holds.

- **(C1) Theory: the order is partial, and no single per-source score survives it** (see Sections 4 and 5). We formalize context sources as statistical experiments on the task's latent requirement, instantiate the Blackwell garbling order with its for-all-tasks dominance guarantee via the Blackwell–Sherman–Stein theorem, define the δ_M-restricted order the experiments probe, and prove a one-way bridging lemma between them. What follows from taking *incomparability* as the object is an impossibility: Blackwell-incomparable sources cannot be ranked by *any* single per-source score $\varphi(W)$, relevance or utility, for all tasks (Theorem 2), with the query-conditioned topical case isolated separately (Proposition 2). Information retrieval has long argued that no single fixed document score is optimal once documents interact (see Section 2.1); we derive that as a theorem in the Blackwell frame and name the exact structural condition under which it bites.
- **(C2) Verification: a measurable deficiency surrogate with finite-sample guarantees** (see Section 6). We define the *population* gap $d_{M,\mathcal{D}}(W_1, W_2) = \max(0, \sup_{T\in\mathcal{D}}[\mathrm{PASS}_M(W_2,T) - \mathrm{PASS}_M(W_1,T)])$ together with its plug-in estimate $\widehat{\delta}_{\mathcal{D}}$, a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform

Table 1: Glossary: the recurring formal terms and their engineering reading.

Formal term	Engineering reading
source W (an experiment)	a candidate context item: a doc, note, or module you could paste into the prompt
Blackwell dominance $W \succeq_B W'$	W answers everything W' answers; keeping W and dropping W' is safe
incomparability	each source is decisive on tasks where the other fails; no single number can rank them once and for all
garbling	W' is a lossy transformation of W: everything in W' is derivable from W
anti-monotonicity ($\Delta_T < 0$)	pasting more true, on-topic text into a working prompt can break it
non-dominated front	the shortlist you must keep whole, because nothing on it verifiably covers anything else on it
verification	a claim measured with exact finite-sample error bars, still significant after multiple-comparison correction, under the per-cell Bernoulli sampling model of Assumption 2
model rule δ_M	the model as it actually behaves on the prompt, not an idealized optimal user of the information
gap $d_{M,\mathcal{D}}$, plug-in $\widehat{\delta}_{\mathcal{D}}$	how many PASS points you lose by using W in place of W' on the task family; the hat marks the sample estimate, never the population gap

verification: exact Clopper–Pearson intervals [9] at Bonferroni level [10] $\alpha = \eta/(2|\mathcal{D}|)$, valid jointly on a single event, so that $L_{\mathcal{D}} > 0$ verifies a positive value gap at confidence $\geq 1 - \eta$. The selection multiplicity is paid explicitly rather than absorbed into a single-sample sup inequality [11], which does not cover an argmax over distinct cells. This is what makes the partial order testable on a real model behind a real verifier.

(C3) Evidence: verified incomparability and scalar-ranking failure on live models (see Section 7). Verified empirical incomparability on six models across four vendors; the ranker-spectrum mis-rank on a confound-controlled trap (lexical cosine, dense bi-encoders, and a cross-encoder all fail; an LLM listwise reranker that reads the decisive content escapes, exactly the topicality boundary the theory draws); and verified anti-monotonicity, a strict superset of a sufficient source collapsing PASS, with order, length-padding, and routing-instruction controls, all with the scope limits stated in Section 8.4.

How to read the formal terms, and where things are. Table 1 gives the engineering reading of each recurring term, and a reader who wants the measured result first can go straight to Section 7. Otherwise the order is prior work (see Section 2), the partial order and the two impossibility results (see Sections 4 and 5), the verification (see Section 6), the six-model measurement (see Section 7), scope and conclusions (see Sections 8 and 9), with proofs, the per-cell record, the diagnostics and the protocol in Appendices A–D.

2 Related Work

2.1 Scalar selection and its limits in information retrieval

Context and demonstration selection is dominated by *scalar* criteria, each inducing a total order and a top-k prefix: topical relevance, lexical or learned [12]; $\mathcal{V}$-usable information [2, 13]; expected information gain and Bayesian experimental design [14, 15]; Shapley- and influence-based valuation of demonstrations [16, 17] and of retrieved documents [18]; perplexity-reduction utility [19]; and binary sufficient-context detection [20]. Each collapses the comparison of two sources to one real number, so each is an instance of the excluded class rather than a competitor: Theorem 2 covers the query-independent ones, and Proposition 2 the topical query-conditioned ones that meet its hypotheses. For the closest neighbour, $\mathcal{V}$-usable information, we prove Blackwell-monotonicity (Proposition 1); for mutual information and fixed-prior EIG it follows from data processing; for a fixed decoder's perplexity-gap we claim nothing (Remark 5). Calibration does not change the class: conformal context filtering [21] calibrates a threshold *on* a scalar and inherits its total order, and scorers of equal ranking quality can retain different document sets [22]. Identifying a non-dominated set from noisy samples is classical in the bandit literature [23]; what is new here is the object it runs on, (model, verifier, task-battery) triples, not the statistical machinery.

Retrieval has itself moved from relevance to *utility* [3], which is model-specific [4] and depends on which other passages are present [24]. That is the same diagnosis from the other side, and this paper is its formal boundary. Utility re-measured per task by running the model is not a fixed per-source score but the probe our estimator is built from; utility *predicted* by a learned scorer is a number again, and falls under whichever of the two results applies. Reading the decisive coordinate is one way out, the listwise boundary; reading public task metadata is another (Remark 6). Three families sit there. Listwise rerankers [25] and learned set selectors [26, 27] are query-conditioned *set* functions, outside the theorem's scope but carrying no dominance guarantee. Skyline selection [7] keeps the non-dominated set, the operator the partial order calls for, but takes dominance over heuristic per-item scores rather than task value. Diversity-aware selection, from MMR [28] to DPPs [29] and submodular selection [30], optimizes similarity geometry rather than verified value. Directed-information γ-covering [31] is closest in form, its covering relation being asymmetric and information-theoretic; but it orders redundancy *between sources*, computed query-agnostically, so it prunes what another chunk predicts and never exhibits two sources each decisive where the other fails.

The limits of scalar relevance are among the oldest themes in the field, and we sit inside that lineage: the Probability Ranking Principle [32] is optimal only under independence, its failure once documents interact is long documented [33], and interdependence was given ranking principles of its own in portfolio theory [34] and the quantum PRP, whose "interference" term names exactly this dependence [35]. The LLM-era version, relevance-maximizing retrieval degrading answers, is measured directly [36], and the observation that relevance does not imply *sufficiency* is argued at the set level [37]. Our delta is the form of the statement: a theorem in the comparison-of-experiments frame, with the structural condition under which every query-independent scalar must mis-rank, and a verified instance on six production models.

2.2 Task-dependence and context degradation

That added context can degrade performance is by now well documented: a single irrelevant sentence tanks reasoning [38]; position matters [39]; high-scoring distractors hurt while random noise can help [40]; retrieval curves are inverted-U in passage count [41]; document count hurts at fixed length [42]; added length alone hurts [43]; and the effect persists across model generations [44] and agent-scaffold components [45]. The genus is older than language models: in data integration, adding sources raises coverage while integration quality first rises and then falls, so that "less is more": low-quality sources deteriorate the fused result rather than bringing the expected gain, and the right move is to select a subset before integrating [5]; the lesson has since been carried explicitly to augmented language models [46]. We do not claim to discover this genus. Two differences locate our finding inside it. Where data integration traces the harm to low-quality sources and restores monotonicity by redesigning the fusion rule, our two sources are individually correct and sufficient, the harm survives gross-length, block-order and formatting controls on the one model where those controls were run and is consistent with convention routing, and the rule δ_M is a fixed model that cannot be redesigned, so the response is selection rather than fusion. And where "interference" in the mechanistic RAG literature means distractor passages steering a model, a failure recently localized and corrected in a sparse feature space [47], and inter-context *conflict* means contexts that contradict each other [48], our sources neither distract nor conflict: both are true and both are needed, and the superset still collapses. Our anti-monotonicity finding contributes the order-theoretic form of this: a strict *superset* of a sufficient source, on sources separately *verified* Blackwell-incomparable, collapses PASS under a per-task verification of the direct contrast $\mathrm{PASS}(W_{1+}, T) < \mathrm{PASS}(W_1, T)$: a reversal no monotone context-value scalar can carry. A representation-level account of when context helps, an additive error decomposition with alignment conditions [49], is complementary: even in its multi-context extension it characterizes *when* a context corrects the error, not an order relation between candidate sources. Pareto-incomparability of prompts along competing metrics *within* a single task [50] is orthogonal to our object, incomparable *sources across tasks*.

2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has entered machine learning and, concurrently with this work, inference-time context selection itself. Alpila [8] formalizes the admission problem against a *frozen decoder* as minimization of Le Cam deficiency, pairs it with the Vereshchagin–Vitányi conditional structure function for the representation problem, proves a capacity floor below which the deficiency cannot be driven, and validates a curve-aware allocator on seven decoders in four families; a plain-language companion states the framing [51] and the artifact is released [52]. Our object is different, the order *between two admitted sources across tasks*: we show it is partial by *measuring* mutual non-domination, derive that no single per-source score can be faithful to it, and verify a strict superset of a sufficient source collapsing PASS, with gross-length, block-order and routing controls run on one model, not every confound excluded. Incomparability in this sense cannot arise inside a single declared decision problem, so its absence there is a different question rather than a gap.

A second line is training-*dataset* valuation. Zheng et al. [53] order datasets by Blackwell informativeness and show that the mutual information between the curated and the test data is a *strategy-proof* score: a curation step that garbles the data, so that $\theta \to D \to f(D)$ is a Markov chain, cannot raise it. That is the data-processing direction, and it is one-directional; a scalar can be monotone along the order without representing it, so their result is compatible with ours rather than opposed to it. Their comparison is also taken against the Bayes-optimal user of the data, ours against a fixed model rule on a task family. What Theorem 2 adds is the converse question, whether any query-independent scalar can be *faithful* to the order, which a monotone score need not be once the order is partial. Nearby uses stop short of what we do: garbling to harden LLM evaluations [54], statistical deficiency to order training *tasks* [55], a deficiency-induced *directional* dominance in transfer learning (no incomparability, no distribution-free bound, outside language modeling) [56], comparison-of-experiments ideas in loss design [57], and a survey of Blackwell's theorems for AI [58]. Another inference-time application sits on the other side of the model: Zhang et al. [59] rank the information structures induced by multi-agent protocols, showing that voting and debate are no more informative than the agents' pooled private information and identifying Bayesian pooled-posterior maximisation as the Blackwell upper bound on any such protocol. That is a dominance statement about how several models' signals combine; ours is an incomparability statement about which single source one model is given, so the two orders are taken over different objects and neither implies the other.

2.4 Where the two results land in the design space

Sorting the methods above on three axes, whether a method is cheap enough to run at retrieval scale, whether it conditions on the task, and whether it can represent a non-dominated *set* rather than a ranking, makes the pattern uniform. Every method cheap enough to precompute emits one number per item, however that number is produced: late interaction over multi-vector representations [60] and retrievers supervised by the language model itself [61, 62] are scored per item no less than lexical relevance is. All of them therefore fall under Theorem 2 when that number is fixed across tasks. The methods that escape do so either by reading the decisive content per candidate, which costs a model call, or by routing on public task metadata, which Proposition 2 does not exclude (Remark 6). No prior family is both cheap and faithful to the partial order; that empty cell is the open problem this paper isolates; the verified non-dominated set pays probe cost to sit in it, and what that cost buys is a verified relation *between* the candidates, not a guarantee about any selector built on the set.

3 Formalization

We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task's latent requirement. This section fixes the objects and separates two decision rules that the informal literature silently conflates: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which every empirical quantity in this paper measures.

3.1 Tasks, latent requirements, and context sources

A *task* T fixes a latent requirement S_T, the object a deterministic verifier v_T checks (e.g. the intended program behaviour). We take S_T to be a *vector of coordinates*

$$S \;=\; (S_1, S_2, \ldots, S_m) \;\in\; \mathcal{S} \;=\; \mathcal{S}_1 \times \cdots \times \mathcal{S}_m, \tag{1}$$

where each coordinate S_j encodes one logically independent requirement. Modelling S as a product of coordinates rather than an opaque atom is what makes the experiment model well posed for fixed-string sources (Section 3.2, Lemma 1).

Definition 1 (Context source as an experiment)**.** A *context source* W is a Markov kernel $\kappa_W : \mathcal{S} \to \Delta(\mathcal{W})$, i.e. a statistical experiment indexed by the latent parameter S, emitting a signal $w \sim \kappa_W(\cdot \mid S)$ in a signal alphabet $\mathcal{W}$ (the source's text). The experiment is identified with its kernel; the sample space is $\mathcal{W}$ and the observed datum is the supplied text w.

Definition 2 (Decision rules and PASS)**.** A *decision rule* is a (possibly randomized) map $\delta : (T, w) \mapsto \widehat{S}$ producing a candidate solution. Utility is the binary verifier $u(\widehat{S}, S) = v_T(\widehat{S}, S) \in \{0, 1\}$, which scores the candidate against the convention in force: the same answer passes under one convention and fails under another, which is why the second argument cannot be dropped. Two decision rules matter:

(i) the *model rule* δ_M realized by a fixed language model M at a fixed decoding configuration θ (model weights and sampling parameters; see Section 3.3). Its realized pass rate on task T under source W is

$$\mathrm{PASS}_M(W,T;s) \;:=\; \mathbb{E}\big[\, v_T(\delta_M(T,w),\, s)\,\big], \qquad w \sim \kappa_W(\cdot \mid s), \tag{2}$$

written $\mathrm{PASS}_M(W,T) := \mathrm{PASS}_M(W,T;S_T)$ at the single realized convention S_T, which is the quantity every measurement in this paper reports;

(ii) the *Bayes rule* $\delta_T^\star$, the utility-optimal rule for task T under a prior μ_T on $\mathcal{S}$ (the task family's distribution over private conventions), achieving the optimal value

$$V^\star(W,T) \;:=\; \sup_{\delta}\; \mathbb{E}_{S\sim\mu_T}\,\mathbb{E}\big[\, v_T(\delta(T,w),\, S)\,\big], \qquad w \sim \kappa_W(\cdot \mid S). \tag{3}$$

Remark 1 (The two quantities are taken under different measures)*.* PASS_M in (2) is *conditional on one latent state*: the expectation runs over the source kernel and the decoder with S held at the realized convention S_T. $V^\star$ in (3) is a *Bayes* value, averaged over μ_T. Consequently $V^\star(W,T) \geq \mathrm{PASS}_M(W,T)$ is *not* automatic from optimality, and we never use it in that form. The obstruction is elementary: let S be a fair bit, let W be uninformative, and let v_T reward naming S; the constant rule "answer 0" has $\mathrm{PASS}_M = 1$ at the state $S = 0$ while $V^\star = 1/2$. Nor can the gap be closed by taking μ_T to be a point mass at S_T: every guessing cap $V^\star(W,T)$ would then equal 1 and the caps of Lemma 2 would lose all content. What optimality does give is the averaged statement $V^\star(W,T) \geq \overline{\mathrm{PASS}}_M(W,T) := \mathbb{E}_{S\sim\mu_T}[\mathrm{PASS}_M(W,T;S)]$. Lemma 2 is therefore proved through the sufficiency of the winning source rather than through any such comparison; the measured pass rates enter its proof nowhere, and neither bound a Bayes value nor estimate a guessing cap.

The distinction in Definition 2 is load-bearing. Blackwell's theorem (see Section 4) is a statement about $V^\star$; every number we report, PASS_M, the deficiency estimator of Section 6, the interference statistic of Section 6.3, and the entire six-model program of Section 7, is a statement about the single fixed rule δ_M. We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*. Reading $\kappa_W(\cdot \mid S)$ as a constant kernel δ_{w_0} independent of S is degenerate: such a kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial σ-algebra, which would make any garbling argument vacuous. We instead model each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

Definition 3 (Projection sources)**.** A source W that reveals the coordinates indexed by a set $A \subseteq \{1,\ldots,m\}$ is the deterministic kernel

$$\kappa_W(\cdot \mid S) \;=\; \delta_{\pi_A(S)}, \qquad \pi_A(S) := (S_j)_{j\in A}, \tag{4}$$

i.e. supplying W lets a decision-maker condition on $\pi_A(S)$ and on nothing outside A. Write $\mathcal{F}_W := \sigma(\pi_A)$ for the sub-σ-algebra of $\mathcal{S}$ generated by the projection. In our instantiation W_1 reveals coordinate S_1 (the call contract, $A_1 = \{1\}$), W_2 reveals S_2 (the wire invariant, $A_2 = \{2\}$), and the superset $W_{1+} = W_2 \mathbin{++} W_1$ reveals both ($A_{1+} = \{1,2\}$).

Under Definition 3 the informativeness of a fixed string is exactly the resolving power of its σ-algebra $\mathcal{F}_W$. The signal alphabet is the range of the projection, and the text "varies with the task" because the realized coordinate values $\pi_A(S_T)$ differ across task instances T. This is what makes the comparison of Section 4 a comparison of experiments rather than a metaphor, and it is what a rigorous treatment of deterministic sources requires [63].

3.3 The conditioning parameter θ

The model rule δ_M carries an implicit parameter θ: the model's weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write $\mathrm{PASS}_M(\texttt{none}, T)$, the model's no-context pass rate, for the part of a private convention that θ already encodes; the value of a source is model-relative through it. When θ already "knows" a convention (the model can guess it without the source), that baseline is high and W's marginal contribution collapses; when the convention is genuinely unknown to θ, the baseline floors and the source has to carry the cell. We deliberately do *not* write this as a conditional mutual information $I(S; W \mid \theta)$: under Definitions 1 and 2 the law of (S, W) is μ_T followed by κ_W and carries no θ, so conditioning on a fixed model index would leave the mutual information unchanged and model-*in*dependent, the opposite of what is measured here; a genuinely model-relative information quantity would need a subjective prior we neither define nor measure. This θ-dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in Section 7, where the guessability of a fixed convention spans 0% to 75% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by θ; we suppress it except where the model-relativity is the point.

4 The Blackwell order over context sources

With sources modeled as experiments (see Section 3), the classical comparison of experiments applies to them directly: we state the order, record that it is partial, and bridge its two registers.

4.1 The order and its two registers

Definition 4 (Garbling order)**.** $W_1 \succeq_B W_2$ (W_1 *Blackwell-dominates* W_2) iff there is a Markov kernel (garbling) $G : \mathcal{W}_1 \to \Delta(\mathcal{W}_2)$ with $\kappa_{W_2} = G \circ \kappa_{W_1}$; i.e. W_2's signal is a noised, post-processed version of W_1's, computable from W_1's signal alone without further access to S.

Theorem 1 (Blackwell–Sherman–Stein; universal dominance for the optimal rule)**.** *Let W_1, W_2 be experiments on a common* finite *parameter space $\mathcal{S}$ with standard Borel signal spaces. The following are equivalent (the implication* (i)⇒(ii) *needs neither finiteness nor further regularity, and is the only direction Corollary 1 and Lemma 2 use):*

- (i) *$W_1 \succeq_B W_2$ (Definition 4);*
- (ii) *for every prior μ on $\mathcal{S}$, every finite decision space, and every bounded loss ℓ, the Bayes risk under W_1 is no larger than under W_2; equivalently the optimal value satisfies $V^\star(W_1, T) \geq V^\star(W_2, T)$ for every task T with bounded utility.*

Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii)*.*

The theorem is classical [1, 63] and we use it as a black box. (i)⇒(ii) holds for arbitrary $\mathcal{S}$ with no further regularity, by simulating the garbling. We invoke the converse (ii)⇒(i) once only, in

Theorem 2, and there for a finite state space, the case Blackwell [1] proves and our coordinate construction realizes; on a general state space the converse needs the convex closure taken in the definition of more-informativeness [64]. Reading "experiment" as a context source (Definition 1) is faithful because the kernels of Definition 3 are standard Borel, which says nothing about the "decision problem" reading: clause (ii) quantifies over *all* bounded utilities, whereas the tasks we can execute are only the binary-verifier subclass $v_T \in \{0, 1\}$ of Definition 2. That restriction is harmless in (i)⇒(ii) but not in the converse, whose witness need not be binary; it costs nothing for the projection sources of Definition 3, whose binary witnesses are exhibited in the proof of Lemma 2(a), and beyond them it is a scope limit of our presentation, not a property of the theorem.

Corollary 1 (Universal context dominance, C1)**.** *If $W_1 \succeq_B W_2$ then, for the* optimal *user, supplying W_1 is at least as good as supplying W_2 on* every *task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information, $\mathcal{V}$-usable information) provides.*

Remark 2 (Preorder on kernels; "partial" means not total)*.* On the class of all experiments $\succeq_B$ is reflexive and transitive but not antisymmetric, since two experiments can garble each other without being equal, so it is a *preorder* and the induced partial order lives on Blackwell-equivalence classes [1]. Throughout, "partial" means *not total*: there exist experiments V, V' with neither $V \succeq_B V'$ nor $V' \succeq_B V$, and on that class the order is not even a lattice [63, 65]. (The bound variables here are not the paper's named sources: our own projection family is better behaved, and the next subsection shows $\succeq_B$ is antisymmetric on it with the induced order the Boolean lattice on coordinate sets, Lemma 1(a).) Incomparability survives on that family and is exactly the regime our experiments target. One caution for what follows: PASS_M is *not* an invariant of a Blackwell-equivalence class, and it need not be: a rule keyed to surface form can prefer differently between two experiments that garble each other. The padding arm W_{pad} of Appendix B reveals no coordinate any verifier checks, so it is Blackwell-equivalent to W_1, and the two score 65% against 50% on `api_post_ok`. That gap is a point estimate, not a verified separation: at $n = 40$ the exact 95% intervals, $[0.48, 0.79]$ and $[0.34, 0.66]$, overlap. The pair illustrates the possibility rather than certifying an instance of it. Either way the δ_M-restricted order of Section 4.3 is not read on the quotient.

4.2 Deterministic sources: incomparability via σ-algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated σ-algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

Lemma 1 (Refinement characterization for projection sources)**.** *Let W_A, W_B be projection sources revealing coordinate sets A, B (Definition 3), with coordinates $S_1, \dots, S_m$ mutually independent under some full-support prior and each S_j non-degenerate. Then:*

(a) $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$*;*

(b) *consequently W_A and W_B are Blackwell-*incomparable *iff $A \not\subseteq B$ and $B \not\subseteq A$, i.e. each reveals at least one coordinate the other hides.*

Lemma 1 is the formal version of the garbling argument used in Section 7: the call contract (W_1, revealing the private arg-order / required-keyword / exception-type facts S_1) and the wire format (W_2, revealing the cents-scale / check-digit facts S_2) give $A_1 = \{1\}$, $A_2 = \{2\}$, and neither set contains the other, so W_1, W_2 are Blackwell-incomparable by construction (and W_{1+}, revealing $\{1, 2\}$, dominates both).

4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns $V^\star$; our experiments measure PASS_M for a single fixed, generally sub-optimal rule δ_M. An inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a δ_M-restricted order.

Definition 5 (δ_M-restricted, $\mathcal{D}$-restricted order)**.** Fix the model rule δ_M and a finite task set $\mathcal{D}$. Say $W_1 \succeq_{M,\mathcal{D}} W_2$ iff $\mathrm{PASS}_M(W_1, T) \geq \mathrm{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$. Call W_1, W_2 $(M, \mathcal{D})$-*incomparable* iff $W_1 \not\succeq_{M,\mathcal{D}} W_2$ and $W_2 \not\succeq_{M,\mathcal{D}} W_1$, i.e. each strictly out-performs the other on some task in $\mathcal{D}$.

$\succeq_{M,\mathcal{D}}$ is the order our verification measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite $\mathcal{D}$); within those restrictions it is still a genuine, *for-all-tasks-in-$\mathcal{D}$* dominance statement, not a scalar.

Lemma 2 (One-way bridge under coordinate separation)**.** *Neither order implies the other unconditionally; each direction of the bridge carries an explicit hypothesis.*

(a) *Let W_1, W_2 be projection sources revealing coordinate sets A_1, A_2 (Definition 3), and let $T_a, T_b \in \mathcal{D}$ be tasks whose verifier-decisive coordinates lie in $A_1 \setminus A_2$ and $A_2 \setminus A_1$ respectively, each verifier being determined by its decisive coordinate together with facts known μ_T-almost surely, and μ_T-almost surely accepting some action measurable in that coordinate and those facts (*sufficiency and solvability under the informed source*; solvability is not implied by the dependence). Let the* guessing caps *$g_a := V^\star(W_2, T_a)$ and $g_b := V^\star(W_1, T_b)$ be the Bayes values* (3) *attainable* without *the decisive coordinate. If $g_a < 1$ and $g_b < 1$, so that neither deprived source determines the coordinate its task turns on, then W_1, W_2 are Blackwell-incomparable. The caps are hypotheses of this criterion, not quantities the experiments determine. Per cell, Section 7 reports a crossover between the same fixed rule's* realized *rates, $\mathrm{PASS}_M(W_1, T_a) > \mathrm{PASS}_M(W_2, T_a)$ and $\mathrm{PASS}_M(W_2, T_b) > \mathrm{PASS}_M(W_1, T_b)$; since PASS_M is state-conditional and $V^\star$ a Bayes value, such a comparison neither estimates nor bounds g_a or g_b, though it is consistent with $g_a, g_b < 1$ (Remark 1); we nowhere report the cap conditions as empirically verified.*

(b) *The converse fails in general: Blackwell incomparability concerns $V^\star$, and a fixed sub-optimal δ_M may fail to realize the optimal-value crossover on $\mathcal{D}$. It holds under the value-attainment hypothesis (Assumption 1).*

Remark 3 (What the caps buy, and what they do not). Neither hypothesis of (a), separation or the caps, can be dropped *from this criterion.* They are not preconditions of structural incomparability as such: for the sources this paper measures, Lemma 1 already separates W_1 from W_2 by coordinate non-refinement, with no caps at all. Lemma 2(a) is a different route to the same conclusion *within the same projection setting*: it argues from Bayes values rather than from σ-algebras, and the caps are what that argument costs. It is not a generalization to arbitrary sources; a criterion for those would be a separate statement we do not prove. What the caps do here is show why a PASS_M crossover alone proves nothing: (W_1, W_{1+}) is $(M, \mathcal{D})$-incomparable on the measured cells (W_1 wins the API cells, W_{1+} the encoding cell) yet strictly Blackwell-*comparable* by construction ($W_{1+} \succeq_B W_1$, the superset revealing one further independent non-degenerate coordinate), precisely because the deprived-side premise fails there: W_{1+} contains W_1's coordinate, so the API-side cap is not small. A bare PASS_M crossover, with neither premise verified, certifies nothing structural in *either* direction. It is equally consistent with genuine incomparability under optimal play (two sources revealing independent coordinates cross even when each side plays optimally) and with a merely incoherent rule (Theorem 3). The two cases should not be run together. (W_1, W_{1+}) is a *strict comparability* whose PASS_M preferences nonetheless oppose; the distinct phenomenon of a rule preferring differently between two *equivalent* experiments is illustrated by the padding pair (W_1, W_{pad}), where the added block carries no coordinate any verifier reads (Remark 2). That a representation-sensitive decoder *can* do this in general is a structural observation, not something our single measured difference certifies. Nor is the projection construction *necessary* for incomparability: it is the cleanest sufficient condition we know, and the one our sources realize.

Assumption 1 (Realized cross-advantage on the witnessing task pair)**.** Let (T_a, T_b) be an *ordered pair of named tasks.* Say δ_M *attains value on* (T_a, T_b) with margin $\gamma > 0$ if $\mathrm{PASS}_M(W_1, T_a) - \mathrm{PASS}_M(W_2, T_a) \geq \gamma$ and $\mathrm{PASS}_M(W_2, T_b) - \mathrm{PASS}_M(W_1, T_b) \geq \gamma$; i.e. the fixed model actually exploits each source's private coordinate on the task that requires it. Despite the historical name we keep for the label, the condition is a statement about *realized* pass rates only: it asserts nothing about the supremum in (3) being attained, and no design can compel a model to satisfy it. The hypothesis is always asserted of an explicitly named pair: it does *not* assert that some such pair exists somewhere in $\mathcal{D}$, and attainment on one pair implies nothing about attainment on another. If δ_M attains value on a pair with $\{T_a, T_b\} \subseteq \mathcal{D}$, then W_1 and W_2 are $(M, \mathcal{D})$-incomparable.

The three objects do different jobs. Lemma 1 establishes structural incomparability for the specified projection sources, with no prior, no caps and no measurement; Lemma 2 is a *conditional* Bayes-value separation criterion whose guessing caps are assumptions, never estimates recovered from the observed state-conditional PASS_M rates; Assumption 1 concerns realized cross-advantage on named tasks, which the experiments assess directly.

5 The incomparability theorem

Section 4 showed the order is partial; we now establish the consequence that motivates the whole paper: no scalar can rank a Blackwell-incomparable pair. We *split* the claim: a theorem for query-*independent* source scores $\varphi(W)$, and a separate, explicitly empirical statement for query-*conditioned* scores $\varphi(W, T)$, the relevance score practitioners actually use; the two must not be conflated.

5.1 Query-independent scores: the theorem

Theorem 2 (No query-independent scalar ranks incomparable sources, C1)**.** *Let $\mathcal{S}$ be finite and let W_1, W_2 be Blackwell-incomparable. (Finiteness is used only to invoke the converse direction of Theorem 1, which turns incomparability into the existence of the witness tasks; it holds for our coordinate construction, and for projection sources the witnesses are exhibited directly in the proof of Lemma 2, so no converse is needed there.) Let $\varphi : \{sources\} \to \mathbb{R}$ be any* query-independent *scalar score (a single number per source: relevance to a fixed corpus, mutual information $I(S; W)$, $\mathcal{V}$-usable information, perplexity-gap, expected information gain under a fixed prior). Being real-valued, φ totally* preorders *sources; it need not order them strictly, since $\varphi(W_1) = \varphi(W_2)$ is permitted. Call φ* faithful *on a task T if the source strictly preferred on T receives the strictly larger φ-value, so a tie is a failure; a task-independent tie-breaking rule fixed in advance reduces ties to the strict case, leaving the conclusion unchanged. Then there exist tasks T_a, T_b such that φ is unfaithful to the optimal-value order on at least one of them, the witnesses being bounded-utility decision problems from Theorem 1*(ii)*, not necessarily binary verifiers (Definition 2), though they can be taken binary for projection sources (proof of Lemma 2*(a)*). If moreover δ_M attains value on* that same *pair (T_a, T_b) (Assumption 1), φ is likewise unfaithful to the realized-*PASS_M *order on at least one of*

T_a, T_b. *More generally, attainment on* any *named pair* (T'_a, T'_b) *yields* PASS_M*-disagreement on at least one of* T'_a, T'_b *by the same argument; the two conclusions concern their own witness pairs and do not transfer between them.*

Remark 4 (Scope). Theorem 2 rules out exactly the class of *source-level* scalars: any φ that is a fixed number per source, regardless of task. $\mathcal{V}$-usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1).

Proposition 1 ($\mathcal{V}$-usable information is Blackwell-monotone)**.** *Let* $\mathcal{V}$*-usable information* $I_{\mathcal{V}}(W \to S)$ *be defined, after Ethayarajh et al. [2], as the reduction in expected predictive risk of the best predictor in a fixed family* $\mathcal{V}$ *when given the signal* w *versus the null signal. If* $W_1 \succeq_B W_2$ *and the predictor family* $\mathcal{V}$ *is closed under pre-composition with measurable garblings, then* $I_{\mathcal{V}}(W_1 \to S) \geq I_{\mathcal{V}}(W_2 \to S)$*. Hence* $I_{\mathcal{V}}$ *is a Blackwell-monotone scalar: it respects* $\succeq_B$ *but collapses it to one real number.*

Remark 5 (Scope of Proposition 1). The closure hypothesis (a family closed under pre-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour, $\mathcal{V}$-usable information. For mutual information and fixed-prior EIG the analogue is not a conjecture but immediate from data processing, the direction Zheng et al. [53] use (see Section 2). A *fixed* decoder's perplexity-gap is the singleton-family case the previous sentence excludes, so we claim nothing for it. Theorem 2 excludes all of them as per-source scalars regardless, which is the only property this paper needs.

5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score $\varphi(W, T)$, e.g. the lexical or embedding relevance that retrieval and RAG actually deploy, induces a *different* order per task and is therefore *not* a single fixed total order, so Theorem 2 does *not* cover it. We make two precise statements instead.

Proposition 2 (Constrained impossibility for feature-measurable scores)**.** *Suppose* $\varphi(W, T) = g(\psi(W, T))$ *for a fixed feature map* ψ *that is* topical*:* ψ *is invariant to the verifier-decisive coordinate of* S_T*. Let* T_c *be a task whose decisive coordinate is supplied by* W_1*, so realized utility ranks* $W_1 \succ W_2$ *on* T_c*. Parts* (a) *and* (b) *are* alternatives*, not a chain: their hypotheses are mutually exclusive. They are not exhaustive of the topical class, and the conclusion below is stated only for scores meeting one of them.* ***(a) (Tie.)*** *If* $\psi(W_1, T_c) = \psi(W_2, T_c)$ *then* $\varphi(W_1, T_c) = \varphi(W_2, T_c)$*, so* φ *cannot rank* W_1 *strictly above* W_2*.* ***(b) (Strict inversion.)*** *Suppose instead that* $\varphi(\cdot, T_c) = g(\mathrm{ov}(\cdot, T_c))$ *reads the pair through the scalar lexical-overlap feature* ov *alone, with* g *strictly increasing, and that* T_c*'s lexicon is saturated with* W_2*'s vocabulary in the precise sense* $\mathrm{ov}(W_2, T_c) > \mathrm{ov}(W_1, T_c)$*. Then* $\varphi(W_2, T_c) > \varphi(W_1, T_c)$*, a strict mis-rank. In either case* φ *fails to rank* $\{W_1, W_2\}$ *by realized utility on* T_c*: no topical score satisfying* (a) *or* (b) *ranks* $\{W_1, W_2\}$ *correctly on all tasks.*

Remark 6 (What Proposition 2 does not exclude). Invariance to the decisive coordinate's *value* is a weak hypothesis, and hypotheses (a) and (b) do not cover every score satisfying it. A rule reading public metadata can escape while never touching a private value. Let $S = (X, Y)$ with X, Y independent, let W_{api} carry the public label `api` and reveal X, let W_{wire} carry the public label `wire` and reveal Y, and let T_{api} and T_{wire} need X and Y. The score $\varphi(W, T) = \mathbf{1}[\,\mathrm{label}(W) = \mathrm{family}(T)\,]$ then ranks correctly on both tasks in every state, and it satisfies neither (a) (the labels differ, so no tie) nor (b) (it is not an increasing function of lexical overlap). So the proposition constrains the two named mechanisms, not the whole query-conditioned class, and in particular it says nothing against routers built on public task metadata. Theorem 2, which concerns one fixed number per source for all tasks, is unaffected: a router of this kind is not a per-source scalar. What the trap shows empirically is narrower: the text-reading scorers we ran put W_2 above W_1 on it (see Section 7.4.2). That is a measured mis-rank, not evidence that a learned checkpoint satisfies (a) or (b); only the lexical cosine meets (b) by construction.

Remark 7 (Why (b) is stated for a scalar overlap score). Strict monotonicity in one *coordinate* of a multi-feature ψ does not give (b). Take $\psi = (\mathrm{ov}, \mathrm{style})$ and $\varphi = \mathrm{ov} + 10\,\mathrm{style}$, which is strictly increasing in overlap. With $\psi(W_1, T_c) = (0.2, 1)$ and $\psi(W_2, T_c) = (0.9, 0)$ the scores are 10.2 and 0.9: overlap favours W_2 while the score favours W_1. Comparing two inputs that differ in other features therefore needs the whole map, which is why (b) assumes φ reads overlap alone. The lexical cosine of Section 7 satisfies that assumption by construction; a learned reranker need not, which is why its behaviour is settled empirically rather than by this proposition.

Because Proposition 2 binds a query-conditioned score only when it meets hypothesis (a) or (b) (Remark 6), we treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use and for the checkpoints of Section 7.4.2. In Section 7 the `trap_store_wire` task realizes T_c, where the lexical cosine and representative open-weight learned rankers—two dense bi-encoders (BGE, E5) and a cross-encoder reranker—all mis-rank the pair (Section 7.4.2, Table 3), so the topical class is not a lexical strawman. The same failure is documented in the wild: language-model re-rankers are steered by lexical similarity across NQ, LitQA2, and DRUID [6], a corpus of real retrieved contexts [66]; and semantically close but logically irrelevant passages defeat retrievers and rerankers at scale, with a listwise evidence-reasoning agent as the remedy [67]. An LLM *listwise* reranker, which reads both documents and can therefore condition on the private coordinate itself, ranks the trap correctly (100/100), behaviour Proposition 2 permits rather than predicts.

6 The decision-restricted deficiency surrogate and its uniform verification

The impossibility of Section 5 concerns the idealized user of a source; every quantity we report, by contrast, comes from a single fixed model. This section builds the instrument that connects the two: a surrogate that measures the order on the fixed rule, and a distribution-free verification that makes each comparison auditable on a live model.

6.1 The surrogate and its correct name

Exact garblings over text cannot be computed, and $\succeq_B$ is only partial, so instead we measure the fixed-rule order of Definition 5. Fix the model rule δ_M and a finite task set $\mathcal{D}$, and define the *decision-restricted value-deficiency*, the worst amount by which W_1 trails W_2 across the tasks in $\mathcal{D}$:

$$d_{M,\mathcal{D}}(W_1, W_2) \;:=\; \max\Big(0,\; \sup_{T \in \mathcal{D}} \left[\mathrm{PASS}_M(W_2, T) - \mathrm{PASS}_M(W_1, T)\right]\Big), \tag{5}$$

where each $\mathrm{PASS}_M(W, T)$ is the *population* pass probability $p_{W,T}$ of (2), so (5) is a population functional and not a statistic; its *plug-in* estimate $\widehat{\delta}_{\mathcal{D}}$ substitutes $\hat{p}_{W,T} = k_{W,T}/n$ from n completions per cell, the hat marking the sample quantity throughout. In population terms $d_{M,\mathcal{D}}(W_1, W_2) = 0$ is *equivalent* to the dominance $W_1 \succeq_{M,\mathcal{D}} W_2$ of Definition 5, and mutual positivity $d_{M,\mathcal{D}}(W_1, W_2) > 0 \wedge d_{M,\mathcal{D}}(W_2, W_1) > 0$ to $(M, \mathcal{D})$-incomparability. Neither holds of the plug-in: at finite n, $\widehat{\delta}_{\mathcal{D}} = 0$

certifies no dominance and $\widehat{\delta}_{\mathcal{D}} > 0$ no positive population gap, both carrying sampling error and the maximum being selected besides. Section 6.2 supplies the missing confidence statement, and that construction, not the sign of the plug-in, is the inferential step on which every claim we call *verified* rests.

Remark 8 (This is *not* Le Cam deficiency). The Le Cam deficiency of $\mathcal{E}$ relative to $\mathcal{F}$, experiments on a common state space $\mathcal{S}$ with kernels $\kappa_{\mathcal{E}}(\cdot \mid s)$, $\kappa_{\mathcal{F}}(\cdot \mid s)$, is the *directed* quantity

$$\delta(\mathcal{E},\mathcal{F}) \;:=\; \inf_{G}\; \sup_{s\in\mathcal{S}}\; \big\| \, G\,\kappa_{\mathcal{E}}(\cdot \mid s) \;-\; \kappa_{\mathcal{F}}(\cdot \mid s) \, \big\|_{\mathrm{TV}}, \qquad \|P-Q\|_{\mathrm{TV}} \;:=\; \sup_{A}\; |P(A)-Q(A)|,$$

with G ranging over Markov kernels from $\mathcal{E}$'s signal space to $\mathcal{F}$'s, so that $\delta(\mathcal{E},\mathcal{F}) = 0$ recovers the garbling relation $\mathcal{E} \succeq_B \mathcal{F}$ of Definition 4 [63, 68, 69]. Its value depends on the direction, on the normalization above (the supremum over events; the L_1 form is twice it), and on $\sup_s$ rather than an average against a prior on $\mathcal{S}$, so we import no constants from the decision-theoretic reading of van Rooyen and Williamson [57], whose uniform excess-risk bound over a *normalized* bounded-loss class is tied to that normalization and unavailable under unrestricted loss scaling. The quantity in (5) performs no garbling minimization and uses a single fixed rule δ_M over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates $d_{M,\mathcal{D}}$ to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (5) the *decision-restricted value-deficiency* and use "Le Cam deficiency" only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative: $d_{M,\mathcal{D}}(W_1, W_2) = 0$ realizes the for-all-tasks-in-$\mathcal{D}$ dominance of Definition 5, the measurable shadow of Corollary 1.

6.2 The uniform verification

The surrogate (5) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

Proposition 3 (Uniform incomparability verification)**.** *Within each cell (W, T) draw n i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let $[L_{W,T}, U_{W,T}]$ be the exact (two-sided) Clopper–Pearson interval [9] for $p_{W,T} = \mathrm{PASS}_M(W, T)$ at level $\alpha = \eta/(2|\mathcal{D}|)$. By the Bonferroni/union bound [10] over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T\in\mathcal{D}}$, all $2|\mathcal{D}|$ intervals hold simultaneously with probability at least $1-\eta$. On that event,*

$$L_{\mathcal{D}}(W_1, W_2) \;:=\; \max_{T\in\mathcal{D}} \big[\, L_{W_2,T} - U_{W_1,T} \,\big] \tag{6}$$

is a $(1-\eta)$ lower confidence bound on $\sup_{T\in\mathcal{D}}[p_{W_2,T} - p_{W_1,T}]$. Hence $L_{\mathcal{D}}(W_1, W_2) > 0$ ***verifies*** *the* population *statement $d_{M,\mathcal{D}}(W_1, W_2) > 0$ at confidence $\geq 1-\eta$. Both directions $L_{\mathcal{D}}(W_1, W_2)$ and $L_{\mathcal{D}}(W_2, W_1)$ read off the* same $2|\mathcal{D}|$ *intervals and so are valid on the same coverage event. Consequently,* if both observed bounds are positive*, the joint incomparability claim $\{d_{M,\mathcal{D}}(W_1, W_2) > 0 \wedge d_{M,\mathcal{D}}(W_2, W_1) > 0\}$ holds at $\geq 1-\eta$ with no additional correction. The condition is not decoration: coverage alone implies nothing about the population gaps, which are identically 0 when W_1 and W_2 are the same source. The verification is one-sided, asserting nothing when the two bounds are not both positive, and the guarantee is frequentist: the probability of declaring incomparability when it does not hold is at most η, which is not a posterior probability that the fixed sources satisfy the inequality. The matching quantity $U_{\mathcal{D}}(W_1, W_2) := \max_{T\in\mathcal{D}} \big[\, U_{W_2,T} - L_{W_1,T} \,\big]$ is, on the same event, a $(1-\eta)$* upper *confidence bound on the same supremum; the dominance control uses it to verify $d_{M,\mathcal{D}}(W_1, W_2) = 0$ up to the equivalence margin.*

A naive single-sample DKW bound under-covers this supremum, because the object is a maximum of differences of binomial means across distinct cells and the verification selects the argmax-gap cell, so the multiplicity must be paid explicitly rather than absorbed into a single-sample envelope (Appendix A); the resulting construction aligns with recent statistical guidance for black-box LLM evaluation (Appendix A).

Assumption 2 (Sampling regime)**.** The n outcomes within a cell are i.i.d. Bernoulli draws of $v_T(\delta_M(T, w))$. Every draw is a single completion (plus deterministic code extraction) from an independent call with no shared state, which is what makes the binomial model of Assumption 2 reasonable rather than proved; we report the decoding configuration in Section 7. Where draws are near-deterministic ($\hat{p} \in \{0, 1\}$ stark cells) the Clopper–Pearson interval becomes one-sided but retains nonzero width, so the verification stays conservative rather than degenerate.

6.3 The interference verification (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source W_{1+} (revealing strictly more coordinates than W_1; Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T \;:=\; \mathrm{PASS}_M(W_{1+}, T) - \mathrm{PASS}_M(W_1, T) - \mathrm{PASS}_M(W_2, T) + \mathrm{PASS}_M(\texttt{none}, T), \tag{7}$$

the second-difference (interaction) of adding W_2 on top of W_1. We write the interaction as Ψ_T (rather than a symbol I that would collide with mutual information $I(S; W)$). A verified $\Delta_T < 0$ on a cell whose W_1 arm is at ceiling is the localized statement that "more strictly-more-informative context hurts"; Ψ_T is reported alongside it as the interaction term only (Remark 9).

Proposition 4 (Interference verification)**.** *Fix a target task $T^\star \in \mathcal{D}$ chosen* before *unblinding, and let $\Delta_{T^\star} := \mathrm{PASS}_M(W_{1+}, T^\star) - \mathrm{PASS}_M(W_1, T^\star)$ be the harm contrast. With exact Clopper–Pearson endpoints $\mathrm{CP}_{\mathrm{lo}}, \mathrm{CP}_{\mathrm{hi}}$ carrying noncoverage at most $\alpha = \eta/2$* per endpoint *on the two arms $\{W_{1+}, W_1\}$ (two endpoints, union-bounding to η; the two unused endpoints of a two-sided interval are not paid for),*

$$\overline{\Delta}_{T^\star} \;:=\; \mathrm{CP}_{\mathrm{hi}}(p_{W_{1+}}) - \mathrm{CP}_{\mathrm{lo}}(p_{W_1}) \tag{8}$$

is a $(1-\eta)$ upper confidence bound on $\Delta_{T^\star}$, and $\overline{\Delta}_{T^\star} \le -\tau$ verifies $\Delta_{T^\star} \le -\tau$ at confidence $\ge 1-\eta$. If instead the harmful cell is selected *from $|\mathcal{D}|$ candidates after unblinding, use $\alpha = \eta/(2|\mathcal{D}|)$; this is the same selection correction the verification applies to the deficiency surrogate (the multiplicity is real: the selected cell is an argmax over $|\mathcal{D}|$ noisy gaps, so a single-sample sup inequality does not cover it), and omitting it would repeat the error we criticize. Because Δ reads only the two arms it contrasts, the bound is insensitive to the* `none` *baseline; we still report that baseline per cell, because guessability bears on whether the source is genuinely private and hence on the incomparability reading (see Section 3.3).*

Remark 9 (Ψ is interaction, not harm). The second difference must not be read as a harm measure. Writing f for the set function $A \mapsto \mathrm{PASS}_M(W_A, T)$, the inequality $\Psi_T \le 0$ is precisely the submodularity condition for the pair $\{W_1, W_2\}$, so a negative Ψ_T is what diminishing returns look like and is consistent with f being monotone submodular. It does not imply that adding W_2 to W_1 hurt: taking $\mathrm{PASS}(\texttt{none}) = 0$ and $\mathrm{PASS}(W_1) = \mathrm{PASS}(W_2) = \mathrm{PASS}(W_{1+}) = 1$ gives $\Psi_T = -1$ with no degradation whatever. What the collapse violates is *monotonicity*, $\Delta_T < 0$, which is the quantity we verify. In our data the two coincide in direction, and the contrast bound is tighter than the four-term one because it spends its confidence budget on two intervals rather than four.

6.4 Anti-monotonicity as information non-monotonicity of δ_M

A verified $\Delta_{T^\star} < 0$ on a ceiling-W_1 cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

Theorem 3 (Information non-monotonicity of the model rule)**.** *Let W_{1+} reveal a superset of W_1's coordinates, so $W_{1+} \succeq_B W_1$ (Lemma 1). Call a rule δ* Blackwell-coherent *on T if its realized pass rate does not fall when it is handed the dominating source, that is* $\mathrm{PASS}_\delta(W_{1+}, T) \geq \mathrm{PASS}_\delta(W_1, T)$ *at the realized state S_T. Coherence in this sense costs no information, and a coherent rule exists on every task: the pruning rule $\delta_M \circ G$, where G deletes W_2's block from W_{1+}'s signal and acts as the identity on W_1's, so that the one rule is defined on both inputs, satisfies* $\mathrm{PASS}_{\delta_M \circ G}(W_{1+}, T) = \mathrm{PASS}_{\delta_M \circ G}(W_1, T) = \mathrm{PASS}_M(W_1, T)$ *at that state, because $G \circ \kappa_{W_{1+}}(\cdot \mid s) = \kappa_{W_1}(\cdot \mid s)$ for every s; no prior and no averaging enter. The Bayes rule $\delta^\star$ is deliberately* not *the exemplar here: its value is monotone in the μ_T-averaged register only, and monotonicity there is consistent with a fall at a single state (Remark 1). If a fixed model rule δ_M has a task $T^\star$ with* $\mathrm{PASS}_M(W_{1+}, T^\star) < \mathrm{PASS}_M(W_1, T^\star)$*, then δ_M is* not *Blackwell-coherent: it violates monotonicity under the garbling order on $T^\star$. In particular no monotone scalar context-value score can predict δ_M's behaviour on $T^\star$.*

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent in the PASS register, and the comparison is like-for-like: δ_M is beaten, at the same state and on the same input, by its own pruned version $\delta_M \circ G$. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.

6.5 Family-wise error across the verified claims

Each verification above controls error *marginally* at η for its own claim (incomparability via Proposition 3; interference via Proposition 4). The unit of accounting is the *coverage event*, not the claim. Every statement that is a deterministic consequence of the same $2|\mathcal{D}|$ Clopper–Pearson intervals holds simultaneously on the event E of Proposition 3, and therefore *jointly* at $\geq 1 - \eta$ with no further correction: this covers both incomparability directions, the dominance control $U_{\mathcal{D}}(\cdot, \cdot)$, whose endpoints are endpoints of those very intervals, and any per-task strict win read off them. Adding a conclusion drawn from intervals already covered does not by itself cost another Bonferroni factor. What does require fresh allocation is a conjunction spanning *different* events, that is, claims reading cells outside E or intervals built at a different level. The interference verification is of that kind, since it reads the W_{1+} and W_1 arms at $\alpha = \eta/2$ per endpoint, and the W_{1+} intervals are not in the W_1/W_2 event at all; conjunctions joining it to the incomparability family are therefore reported as marginal per claim, and where a joint statement is wanted we allocate η across the C events involved.

The two verifications have different effective-n requirements: incomparability is carried by stark 0/1 cells whose intervals are tight, whereas a harm contrast on a milder collapse spends more of the budget. This is why the four-term interaction bound left Opus short of the margin while the two-arm contrast clears it (see Section 7.4.3); the power analysis is in Appendix A.

7 Experiments

With the order (see Section 4), its impossibility consequence (see Section 5), and the verification that measures it (see Section 6) in hand, we now instantiate the theory on real language models. The design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (see Section 7.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every verified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (see Section 7.4), so the verification is auditable rather than asserted.

7.1 Design: two incomparable context sources

We define two *private, unguessable* conventions on the *same* module ledger, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module, same task shape). Because both are on-topic for any "ledger" task, any lexical, embedding, mutual-information, or $\mathcal{V}$-usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

W_1, **API call contract.** ledger.post(account, amount, *, memo): account first, amount second positional, memo a *required keyword-only* argument; returns an integer entry id ≥ 1; raises ledger.PostError (not ValueError/KeyError) on failure.

W_2, **wire-encoding invariant.** Amounts serialize to <sign><minor-unit-digits>#<check>, where check = (sum of digits) mod 10; e.g. \$12.30 $\mapsto$ "+1230#6" and $-$\$0.07 $\mapsto$ "-7#7". Decoding rejects a wrong check digit.

W_{1+}, **literal superset.** $W_{1+} = W_2 \mathbin{++} W_1$, the concatenation of W_2's signal with W_1's. As a deterministic source revealing both coordinates, it Blackwell-dominates both W_1 and W_2 ($W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$); it is the anti-monotonicity / dominance probe. We also use $W_{1+}^{\mathrm{rev}} = W_1 \mathbin{++} W_2$ (W_1 first) as an order control.

Why incomparable (garbling argument). The argument is simple. From W_2's wire format you cannot recover W_1's private argument order, its required memo keyword, or its error type; from W_1's contract you cannot recover W_2's cents scale, its # separator, or its mod-10 check. Each source states a fact the other leaves out, so neither is a blurred copy of the other, and W_1, W_2 are incomparable. This is exactly the coordinate picture of Section 3: W_1 and W_2 reveal different, non-overlapping facts about S (Lemma 1).

Confound defense: the double crossover. The key pattern is a *double crossover*, measured against a none baseline (no note supplied): W_1 raises the API tasks but *not* the encoding task, while W_2 raises the encoding task but *not* the API tasks. If encoding were simply "harder," W_1 would fail to help it and W_2 would still have to raise it from the same floor. In the clean carrying cells the deprived arms floor at or near the none baseline, so the two task families are equally hard on their own and any difference comes from the matching private fact, not from one family being harder; when a model prior leaks a convention (Table A1), we report that leakage explicitly and do not use those cells for claims that require a near-zero baseline.

7.2 Tasks

The full design comprises seven tasks with stub-ledger verifiers (return code 0 = PASS): four W_1-decisive API tasks (api_post_ok, api_argorder, api_error_type, api_memo_required), two

W_2-decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

Verified battery. The verified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the verified battery. The four-cell selection was made to retain at least one W_1-decisive, one W_2-decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (see Section 7.4.1).

The relevance trap. The trap `trap_store_wire` is a constructed test of the topicality condition in Proposition 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks $W_2 > W_1$; but its graded requirement is the W_1 call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks $W_1 > W_2$. This defeats the query-conditioned relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine $\varphi_{\text{lex}}(W, T)$ between a source and the task prompt, computed *before* the model is run. We use lexical cosine as the *minimal* instantiation of Proposition 2, the cheapest member of the topical class; stronger learned rankers (dense bi-encoders, a cross-encoder, and an LLM listwise reranker) are tested against the same trap separately in Section 7.4.2, while the theorem (Theorem 2) is the general claim for query-independent scalars.

7.3 Materials and Methods

Models and vendors. We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6. Nothing in this paper is trained or fine-tuned: all six are fixed production systems accessed through their public APIs, so there are no training or validation curves to report; the released measurement record, per-cell PASS tables, exact intervals, and raw run logs, is the complete empirical object.

Harness and arms. The harness (`harness/measure_blackwell.py`; released with all run logs at https://github.com/abilous-ti/blackwell-llm-context) is Python-standard-library only (zero external dependencies); it computes $\widehat{\delta}_{\mathcal{D}}$ in both directions, the uniform Clopper–Pearson/Bonferroni verification, the interference verification, and the lexical relevance score, and it implements the launchers for the Anthropic Messages API, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms $\{\texttt{none}, W_1, W_2, W_{1+}\}$. PASS is measured over n completions per cell.

One turn budget, one transport. Every model gets a single completion per attempt, from which we extract the candidate code and grade it. All six are reached the same way: a single HTTP request the harness composes itself, carrying only the task prompt, with no system prompt of our own and no tools available to the model. The Anthropic models are served over the Messages API and the other three over an OpenAI-style endpoint, but in both cases the request is one pass with no agentic exploration, so the model must use what it is given. Cross-vendor *magnitudes* therefore no longer carry a transport confound, though they remain subject to the ordinary caution that vendors' models differ in ways we do not control (see Section 8.4). The claims the paper verifies are within-model comparisons between arms, where any remaining request-level detail is held fixed and cancels.

Statistical reporting. Each cell is $n = 40$ independent completions scored by an executable verifier and summarised as a count; the arms within a cell are separate completions, and each draw is an independent call with no shared state (Assumption 2). Uncertainty is carried by exact Clopper–Pearson intervals rather than a normal approximation, which at $n = 40$ and proportions near 0 or 1 is where the approximation is worst (see Section A). Multiplicity is paid explicitly by a union bound whose family is stated with each claim, at confidence budget $\eta = 0.10$, so the statements *within* a family hold jointly with probability $\geq 1 - \eta$. Incomparability spends $\alpha = \eta/(2|\mathcal{D}|)$ over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T\in\mathcal{D}}$, and both directions are read off those same intervals (Proposition 3); a harm contrast spends $\alpha = \eta/2$ on its two arms $\{W_{1+}, W_1\}$ and verifies $\Delta_T \leq -\tau$ at $\tau = 0.30$ through the upper bound $\overline{\Delta}_T$ (Proposition 4), the four-term Ψ_T being reported descriptively only (Remark 9). Each verification is thus controlled marginally at η; we do not claim joint control across the incomparability, dominance and interference families drawn from a single run (see Sections 6.5 and 8.4). At $\hat{p} = 0$ or $\hat{p} = 1$ the interval becomes one-sided but does not degenerate to a point: at $n = 40$ and the level Figure 1 uses, 0/40 gives [0, 0.128] and 40/40 gives [0.872, 1]. It is that residual width a crossing must beat, which is why the stark 100%/0% cells verify at this n while intermediate proportions would need a larger one. Results whose intervals do not separate are labeled *reproduced but not verified* and are never used as evidence; the protocol reports its own insufficiency rather than hiding it. We report no p-values and fit no model to the data.

Reproducibility artifacts. The measurement harness, every per-cell run log, every result file and the analysis scripts are released (Data Availability Statement). `results/MANIFEST.md` pins each record by SHA-256 and maps every table and figure to the files it is computed from, and `harness/diag/check_regimes.py` reports the transport and turn budget of each cited run with the evidence for it. It is an inventory, not a proof: it checks only the runs on a fixed list, and matches the disclosure wording against the manuscript globally rather than to the run it describes. Appendix D gives the decoding configuration, the retry and outage policy, and the cells that were re-measured, with why.

Why these constants. Both constants were fixed before the runs and neither is load-bearing. $\eta = 0.10$ is a conventional 90% family level, and the headline separations are near-0% against near-100%, so tightening it does not move the verified conclusions. $\tau = 0.30$ is an a priori materiality threshold: to count, interference must destroy at least 30 PASS points, well above the interval widths at $n = 40$, so noise cannot clear it. The data show the threshold doing real work. Verified collapses clear it with upper bounds from -0.38 to -0.85 (Table 4) and the character-matched padding control at -0.67. What shows the choice is not load-bearing is that the verdict barely moves with it: every one of the twelve contrasts clears marginally for any τ from 0.10 to 0.30, and eleven of twelve up to 0.50, so halving or raising the threshold leaves the conclusion intact. The margin is set by DeepSeek's two cells, at -0.385 and -0.541, which are the first to drop out as τ rises.

What the intervals do not cover. The Clopper–Pearson intervals carry sampling error under Assumption 2, a fixed success probability per cell. They carry nothing else, and two components

Table 2: Consolidated 6-model / 4-vendor verdict. Every model gets one completion per attempt at $n = 40$ per cell; the transport differs between the Anthropic models and the rest, so cross-vendor magnitudes are compared under one uniform protocol (see Section 7.3), so what separates vendors here is the models, not the request path. "Anti-mono. verified" counts, of the two API cells tested per model, those on which the harm contrast's upper bound clears $-\tau = -0.30$ at $\eta = 0.10$. "Argorder `none`" is the no-context baseline PASS on `api_argorder`, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers. Confidence bounds are rounded outward (a lower bound down, an upper bound toward zero), so every printed bound is implied by the computed one.

Claim	Haiku	Sonnet	Opus	GPT-5.5	DeepSeek	Kimi-K2.6
Vendor	Anthropic	Anthropic	Anthropic	OpenAI	DeepSeek	Moonshot
n per cell	40	40	40	40	40	40
Incomp. verified ($\geq 90\%$)	✓	✓	✓	✓	✓	✓
$L_{\mathcal{D}}(W_1, W_2)$	+0.76	+0.71	+0.71	+0.76	+0.34	+0.67
$L_{\mathcal{D}}(W_2, W_1)$	+0.76	+0.76	+0.76	+0.76	+0.71	+0.76
Relevance mis-rank (trap)	✓	✓	✓	✓	✓	✓
Anti-mono. verified (# cells, of 2)	1	2	2	2	2	2
`api_argorder none`	2%	0%	40%	75%	2%	45%

outside them are larger than they are. *Model drift*: successive re-measurements moved Opus's `api_post_ok` superset arm from 38% to 12% and then to 3%, and DeepSeek's W_1 arm from 75% to 43%, both far outside the nominal intervals for those cells. *Prompt wording*: an earlier record traced a 100% → 52% swing on one cell to the prompt naming the call as `ledger.post`, an effect an order of magnitude wider than any interval here, and each cell has exactly one prompt, so this component is not modelled at all. Every verified quantity is therefore an inference about a (model snapshot, prompt, default decoder, collection window) tuple, indexed by θ (see Section 3.3) and by the run it was drawn in, not a property of a vendor's model line. We read that as a weakness: the re-measurement bounds the drift component empirically, and the direction of every verdict survived it.

Provenance. Four re-measured arms or cells sit in grids whose other arms come from earlier runs (Appendix D): the Opus W_{1+} arm, carrying both Opus rows of Table 4; the Opus W_2/`trap_store_wire` and Haiku W_1/`enc_amount` cells, the latter carrying one of Haiku's two incomparability directions; and the structured-context arm of Appendix B. Those contrasts are across runs: a shared model name and transport do not establish a shared decoder, and their intervals price sampling error within an arm, not the gap between windows. The six-model record is itself contemporaneous and retained (Appendix C.1), so the harm contrasts are read within one window; `results/MANIFEST.md` gives the run, and so the window, behind each number.

7.4 Results

We report four findings: incomparability (see Section 7.4.1), the relevance mis-rank (see Section 7.4.2), anti-monotonicity (see Section 7.4.3), and the model-relative value of private context (see Section 7.4.4). Table 2 consolidates the per-model verdicts; Tables 4 and A1 give the underlying quantities.

7.4.1 Incomparability verified on all six models

On every model the empirical deficiency is mutually positive, $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ and $\widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$, and the uniform lower bounds $L_{\mathcal{D}}$ in both directions clear zero, verifying empirical incomparability at $\geq 90\%$ (Table 2). The crossover is stark: W_1 solves the API and trap cells and fails encoding, while W_2 solves encoding and fails the API and trap cells. For Haiku at $n = 40$ the carrying cells are `api_argorder` (`none` 2% / W_1 100% / W_2 0%) and `enc_amount` (`none` 0% / W_1 0% / W_2 100%), giving $L_{\mathcal{D}} = +0.762/+0.762$, far clear of zero.

The six-model claim, made jointly. Proposition 3 controls the $2|\mathcal{D}|$ intervals of *one* model's run, so the per-model verifications above are each at $\geq 1 - \eta$ and their conjunction across the roster is not. Rather than leave the six-model statement marginal we pay for it: allocating η across the six runs, i.e. per-*interval* level $\eta/(2|\mathcal{D}| \cdot 6) = \eta/48$, which places $\eta/96$ in each tail, every one of the twelve directions remains strictly positive, binding at DeepSeek-V4-Pro's $L_{\mathcal{D}}(W_1, W_2) = +0.263$ and reaching $+0.684$ on the stark cells. The claim "all six models are empirically incomparable" therefore holds *simultaneously* at $\geq 90\%$, not merely model by model. Table 2 reports the per-model bounds; the simultaneous ones are uniformly smaller by 0.077 to 0.084 and change no verdict.

Because incomparability is a two-directional claim, both $L_{\mathcal{D}}$ bounds must be positive; the smaller of the two is the binding one. Two cells carry each model's verification, and which two is model-specific: on five models they are one API cell and the encoding cell, while on DeepSeek-V4-Pro the W_1-over-W_2 direction is carried by the *trap* cell rather than an API cell (its API margins being smaller). The effective support of the surrogate is therefore two cells per model, drawn from a battery of four. We therefore describe $\mathcal{D}$ as a fixed adversarial four-cell battery rather than a sampled task distribution: the verification is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition run against six models—six rules on one pair, with W_1, W_2 and $\mathcal{D}$ held fixed and only δ_M varying, not six independent source-pair replications. The n draws behind a cell are repeated outputs on *one* task, which is what the Bernoulli model of Assumption 2 prices; no interval in this paper samples tasks.

7.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$, so it prefers W_2; yet realized PASS ranks $W_1 \gg W_2$ on all six models (Haiku 92% vs. 0%; Sonnet, Opus, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%). On the two non-trap cells relevance *agrees* with PASS, on the API cells relevance ranks $W_1 > W_2$ and so does PASS, and on the encoding cell relevance ranks $W_2 > W_1$ and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The `none` floor on the trap is 0% on all six models, and W_2 scores 0% there too, confirming both that the convention is unguessable and that encoding knowledge is useless on this task.

The ranker spectrum is the point here: dense retrievers and a cross-encoder fall for the trap while a listwise reranker escapes it. To test whether the mis-rank extends beyond bag-of-words, we scored $\{W_1, W_2\}$ against the probe tasks with representative open-weight rankers, two dense bi-encoders (`bge-small-en-v1.5`, `e5-small-v2`) and a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`), and with a RankGPT-style LLM *listwise* reranker (Haiku one turn, seeing the task and *both* sources; both presentation orders; $n = 100$ judgments per task). Table 3 reports the induced rankings. Every feature-based ranker we tested mis-ranks the trap: both dense bi-encoders prefer W_2 on the trap while ranking both non-trap cells correctly, textbook topical behaviour, and the cross-encoder prefers W_2 on the trap with maximal confidence (logits -5.63 for W_1 vs $+3.24$ for W_2) while

Table 3: The ranker spectrum on the probe tasks. Each cell is the ranker's top pick between $\{W_1, W_2\}$; "PASS" is the realized-utility ranking from the verified record. Dense and cross-encoder scores are deterministic; the LLM listwise row pools $n = 100$ judgments per task (unanimous, both presentation orders). $\times$ marks a mis-rank. The cross-encoder's `enc_amount` pick is a near-tie (-0.279 vs -0.305).

Ranker	Family	`trap` (PASS: W_1)	`api_post_ok` (PASS: W_1)	`enc_amount` (PASS: W_2)
bag-of-words cosine	lexical	W_2 $\times$	W_1	W_2
`bge-small-en-v1.5`	dense bi-encoder	W_2 $\times$	W_1	W_2
`e5-small-v2`	dense bi-encoder	W_2 $\times$	W_1	W_2
`ms-marco-MiniLM-L-6-v2`	cross-encoder	W_2 $\times$	W_1	W_1 $\times$
Haiku listwise (n=100)	LLM listwise	W_1	W_1	W_2

being near-tied (and narrowly wrong) on the encoding cell. The LLM listwise reranker, which can read the verifier-decisive coordinate rather than match surface features, tracks realized PASS everywhere, unanimously: 100/100 per task, split evenly across the two presentation orders so that a ranker merely preferring the first document would show. Unanimity at $n = 100$ puts the exact 95% Clopper–Pearson lower bound on the preference rate at 0.964; at the pre-declared halfway checkpoint of 50 judgments it was 0.929, and 10 judgments would have bounded it only at 0.692, too close to chance for a claim about consistent ranking. This sits on the boundary Proposition 2 draws: its hypothesis is *topicality*, invariance to the decisive coordinate, and the trap defeats the lexical, dense and cross-encoder checkpoints we ran while sparing the one ranker that reads content. Two limits on how far this generalizes. The three families are three checkpoints on one trap, so their failure is measured, not proved for the families at large; and the proposition does not establish that a model call per candidate is *necessary* to escape, since a router reading public task metadata would also escape at no such cost (Remark 6). What the measurement does show is that the scorers actually deployed here, which read text, pay that price.

7.4.3 Anti-monotonicity: a verified superset that hurts

The superset W_{1+}, which literally *contains* W_1's signal, sharply degrades the API cells relative to W_1 alone. The per-task harm contrast $\Delta_T = \mathrm{PASS}(W_{1+}, T) - \mathrm{PASS}(W_1, T)$ is verified negative ($\Delta_T \le -\tau$, $\tau = 0.30$) on all twelve API cells, two per model, and so on every one of the six models (Table 4, Figure 2). Those twelve are *marginal* verifications, each at $\ge 1 - \eta$ for its own cell; the conjunction can be bought from the same records, since spreading η across all twelve contrasts, a per-interval level of $\eta/12$ over the 24 endpoints, still leaves eleven clear of $-\tau$ with every model keeping at least one. The roster-level statement "on each of these six models some strict superset verifiably harms some cell" therefore holds *simultaneously* at $\ge 90\%$, not merely model by model. The mildest verified collapse is DeepSeek's `api_argorder`, 68% $\to$ 5% at upper bound -0.38, one of the two cells the joint budget drops; all twelve also survive a Bonferroni correction across the two cells within each model. Haiku's shallowest collapse is on `api_argorder`, 98% $\to$ 5%. In the superseded command-line record that same cell was depressed on the W_1 side instead, by an omitted `import ledger` line rather than by the contract (see Section 7.4.5). This is a verified empirical falsification of superset-*monotonicity* for these fixed rules (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it. Table A1 gives the same collapse arm by arm for every model.

The idealized Blackwell order on $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ collapses the choice to the superset: W_{1+} reveals both coordinates, so $W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$, and "more signal is better" keeps $\{W_{1+}\}$

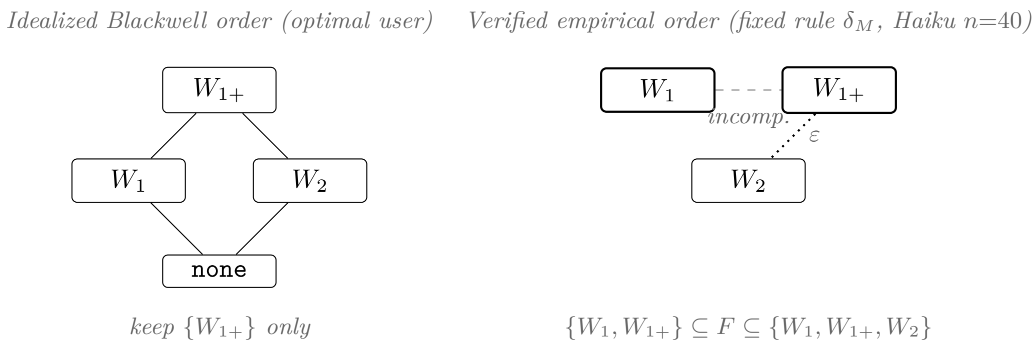

Figure 1: Two orders over the same three sources, $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ with $W_{1+} = W_2 +\!\!+ W_1$. Left: the Blackwell order for the optimal user, where the superset reveals both coordinates and dominates. Right: the order realized by the fixed model rule, on Haiku at $n = 40$, read off a single coverage event over the 3×4 Clopper–Pearson intervals (text above). A dashed edge is a verified incomparable pair; a dotted edge is an ε-dominance only, at $\varepsilon = 0.1281$, so W_2 cannot be excluded exactly. The `none` arm is drawn only on the left, since it is outside that coverage family and no relation to it is verified. The right-hand panel is therefore a verified *partial* picture rather than a determined Hasse diagram: what it establishes is $\{W_1, W_{1+}\} \subseteq F \subseteq \{W_1, W_{1+}, W_2\}$.

alone. The verified empirical relation for the fixed rule has the opposite shape (Figure 1): W_1 beats W_{1+} on the API cells, W_{1+} beats W_1 on the encoding cell, and W_1 and W_2 are the verified incomparable pair, so the *point estimate* of the PASS_M-non-dominated front is the two-element set $\{W_1, W_{1+}\}$, an object no scalar top-1 rule can represent; the two-sided inclusion below bounds it. Three sources over four cells need *one* coverage event, so all 3×4 Clopper–Pearson intervals are taken at per-interval level $\eta/12$; they hold jointly at $\geq 1 - \eta$ and every relation in the figure is read off that single event. On the n=40 Haiku record $L_{\mathcal{D}}(W_1, W_2) = +0.2070$ and $L_{\mathcal{D}}(W_2, W_1) = +0.7439$, both positive, which verifies the incomparable pair. Non-dominance is likewise a *lower*-bound statement, and the same event supplies it: $L_{\mathcal{D}}(W_{1+}, W_1) = +0.7439$, witnessed by `api_argorder`, shows that W_1 beats W_{1+} on some cell, and $L_{\mathcal{D}}(W_1, W_{1+}) = +0.7439$ shows the converse, so neither dominates the other. An upper bound could carry neither claim: it bounds how large an advantage *might* be, not that one exists. The relation the front turns on has only an upper bound, $U_{\mathcal{D}}(W_{1+}, W_2) = 0.1281$. That is an ε-dominance rather than an exact one, saying that on *every* cell of the battery W_2 can beat W_{1+} by at most 0.1281, so excluding W_2 is licensed only up to that margin. What the coverage event establishes exactly is the two-sided inclusion

$$\{W_1, W_{1+}\} \;\subseteq\; F \;\subseteq\; \{W_1, W_{1+}, W_2\},$$

where F is the PASS_M-non-dominated front on $\mathcal{C}$: both W_1 and W_{1+} are verifiably in it, and at $n = 40$ a three-element front cannot be ruled out. The selection rule the figure implies, keep the non-dominated set rather than force the superset, holds under either reading.

What the position controls show. On Haiku, the collapse persists under reversal of the two source blocks, and character-matched off-topic padding does not reproduce it. The padding control (Appendix B) fills the slot W_2 occupies inside W_{1+}, W_1 last in both, and leaves W_1 at ceiling on `api_argorder`: changing only that slot's content turns a harmless arm harmful. The reversed arm $W_{1+}^{\text{rev}} = W_1 +\!\!+ W_2$ puts W_1 first, over the same transport and $n = 40$ as the main table (`blackwell_controls_api_n40`). The two orderings are indistinguishable on both cells: W_{1+} and

Table 4: Anti-monotonicity verification on the carrying cell per model. $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$ is the harm contrast; $\overline{\Delta}_T$ is its $\eta/2$ Clopper–Pearson upper bound over the two arms, and the cell is verified harmful when $\overline{\Delta}_T \leq -0.30$. The $W_1 \to W_{1+}$ column reports the PASS collapse. Unlike the four-term interaction Ψ_T, the contrast does not read the `none` arm, so a leaky baseline cannot slacken it.

Model	Carrying cell	$W_1 \to W_{1+}$	Δ_T	$\overline{\Delta}_T$	Verified
Haiku-4.5	`api_argorder`	80% → 0%	−0.80	−0.59	✓
Haiku-4.5	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
Sonnet-4.6	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
Sonnet-4.6	`api_argorder`	100% → 0%	−1.00	−0.85	✓
GPT-5.5	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
GPT-5.5	`api_argorder`	100% → 0%	−1.00	−0.85	✓
DeepSeek-V4-Pro	`api_post_ok`	75% → 0%	−0.75	−0.54	✓
DeepSeek-V4-Pro	`api_argorder`	68% → 5%	−0.62	−0.38	✓
Kimi-K2.6	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
Kimi-K2.6	`api_argorder`	100% → 0%	−1.00	−0.85	✓
Opus-4.8	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
Opus-4.8	`api_argorder`	100% → 0%	−1.00	−0.85	✓

W_{1+}^{rev} each take W_1 from 98% to 0% on `api_post_ok` and from 78% to 0% on `api_argorder`, verified harmful at −0.81 and −0.57 respectively. Moving the added block out of the primacy slot therefore changes neither the sign nor the size of the harm, which the collapse. On one model and in this configuration, the controls constrain positional and gross-length explanations without identifying a unique cause or isolating tokenization, processing difficulty, or the internal mechanism; the reversed arm bounds the *existence* of a positional explanation, not position's contribution to the harm's size.

Robust across the capability ladder. The harm is large and robust across the Anthropic capability ladder. On the two API cells the $W_1 \to W_{1+}$ collapse is Haiku 100%/80% → 0%/0%, Sonnet 100% → 0%/0%, and Opus 100% → 0%/0% (Table 4). Both stronger models sit at the floor on both cells, so this record orders neither for nor against a capability trend: an earlier two-point reading suggested the harm "intensifies with strength", and we now claim only robustness across the ladder, not a capability trend. All three Anthropic models collapse to 0% on both cells, so there is no residual spread to order. What survives is the negative claim, not the ranking that motivated it.

7.4.4 The model-relative value of private context

The "private" conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The `none`-arm baseline on `api_argorder` forms a clean spectrum: Sonnet and Haiku 0%, DeepSeek 2%, Kimi 45%, GPT-5.5 75%. Opus is excluded from this comparison: it solves `api_argorder` unaided in 40/40 draws, so for that model the cell measures no value of private context at all (Table 2). So the value of the private context, the model-relative guessability $\text{PASS}_M(\texttt{none}, T)$ of Section 3.3, varies continuously across models, and it is not ordered by capability. What it does track we cannot identify from these runs. DeepSeek, a different vendor, has a tight prior like the smaller Claude models, while GPT-5.5's prior already holds much of the convention (`api_argorder` 75%, `enc_amount` 55%), so a "stronger-model-knows-more" account does not fit. Utility-centric retrieval reaches the same conclusion from the other side, the same passage

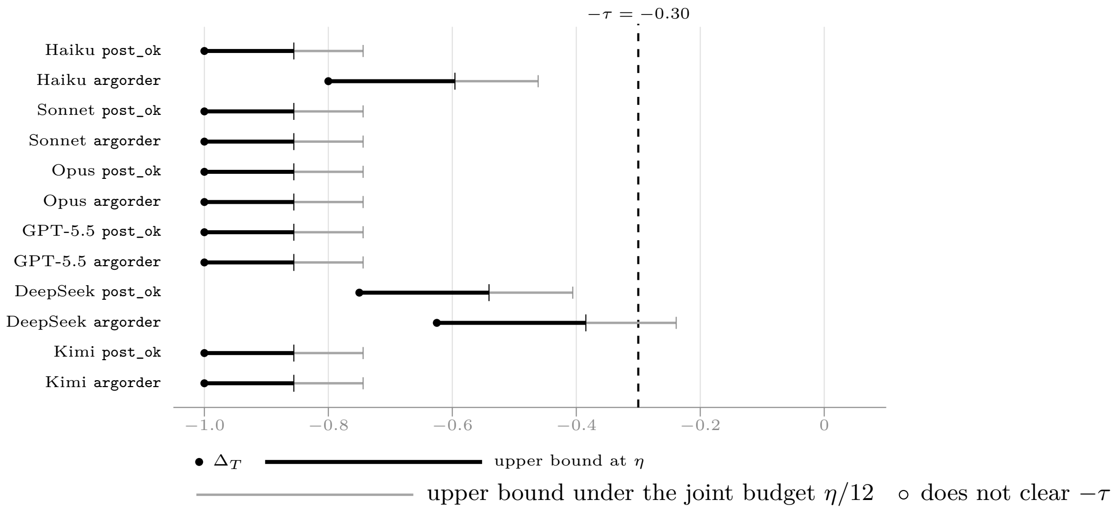

Figure 2: The harm contrast $\Delta_T = \mathrm{PASS}_M(W_{1+}, T) - \mathrm{PASS}_M(W_1, T)$ on both API cells of every model, with the bound that carries the verification. A cell is verified harmful when the *upper* bound clears $-\tau = -0.30$, so a bar ending left of the dashed line is a verified collapse. Thick bars are the marginal bounds at $\eta = 0.10$, one family per cell: all twelve clear. Thin grey bars are the same contrasts under a single η spread across all twelve, at per-interval level $\eta/12$: eleven clear, and every model keeps at least one. Values are those of Table 4, recomputed from the released per-cell counts.

carrying different utility for different models [4]; what the spectrum adds is the verified per-model floor. Within Anthropic the ordering is not by capability either: Sonnet guesses least (0%) while the stronger Opus guesses more (40%). The verified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok` `none` = 5%, trap = 0%).

7.4.5 What the failing replies contain

Every draw behind the interference table was kept, 80 completions per arm and cell on Haiku, and read directly (Appendix C). Three things follow, and one does not. *Packaging is not the explanation*: the extractor returned a non-empty code block on every retained draw, and the `import ledger` line that a command-line record of this cell was missing about half the time is present in 80/80 of the W_1 replies and 78/80 of the superset replies, so neither an empty extraction nor a missing import accounts for the collapse here. *The harmful arm is verbose*: mean reply length runs 139 characters against 509 on `api_post_ok` and 466 against 713 on `api_argorder`, the same inflation reported in Section 8.3. *No reply declined the task*: refusal language appears in none of the 320 retained Haiku completions.

What the retained text does *not* establish is the mechanism. A strict test for the failure mode an earlier command-line record showed, the superset reply applying W_2's wire encoding to a plain-cents amount, fires on 0/80 and 1/80 of the failing superset replies here. So on this record the packaging explanations are excluded and the collapse is not attributed to any internal cause; Theorem 3 constrains any rule whose PASS falls under a strict superset, whatever that cause turns out to be.

7.5 Generality and controls (summary)

Four follow-up experiments address three threat axes (source-pair generality, the context-length confound, and prompt-curability of the collapse); full protocols and numbers are in Appendix B. **(i)**

A second incomparable pair, different domain: a private `cache`-module contract versus a key-normalization invariant is again verified incomparable on Haiku at $n = 40$ ($L_{\mathcal{D}} = +0.728/+0.586$), over the same transport as the six-model record. The superset harms there too but mildly, taking V_1 from 72% to 65% on the trap cell, far short of the $\tau = 0.30$ margin, so the second domain carries the crossover rather than a second verified collapse (Section 8.4). **(ii) Length-matched padding**: filler character-matched to W_{1+} leaves W_1 at ceiling on both collapse cells (100% against W_{1+}'s 0%), which constrains the gross-length explanation on this model. **(iii) Routing instruction**: a one-line convention-routing note recovers most of the collapse (0% $\rightarrow$ 82% and 0% $\rightarrow$ 98%); writing that note requires knowing which source is decisive for which task family, which is exactly the coordinate structure the partial order formalizes, so the cure presupposes the diagnosis. **(iv) Structured context**: wrapping the same two conventions in labeled XML sections *without* any routing statement does not mitigate at all (0% on both cells, against 82% and 98% with the routing note), so segmentation alone does not cure the interference and the operative variable is routing knowledge. The mitigation hierarchy, raw superset < structure < routing < dropping the dominated signal, is itself a finding: it locates the failure in convention-to-task routing rather than in convention segmentation.

7.6 The engineering consequence: four selection policies, one measured table

Table 5 scores four selection policies per model as mean PASS over the four verified cells of Table A1. *Best fixed source* grants the entire query-independent-scalar class its best case: any such scalar induces one total order, hence one top-1 reused for every task, and we let it pick the oracle-best source (W_1 for every model). *Topical top-1* re-scores per task with topical relevance, choosing W_1 on the API cells and W_2 on encoding, but also W_2 on the trap, where every measured topical ranker puts the wrong source first (see Section 7.4.2). *Concatenate both* refuses to choose and submits W_{1+}. *Oracle over* $\{W_1, W_2\}$ takes the per-cell maximum over the non-dominated pair. It is scored on the same outcomes it maximizes over, so it is an oracle and not an achieved result: a deployable router would have to be fixed before testing and evaluated on held-out tasks under matched latency and token budgets, which this paper does not do. That column is selection *headroom* on this battery, not a delivered gain: the oracle's 95.10% mean against 71.72% (best fixed source), 70.52% (topical top-1) and 46.56% (concatenation).

Three structural facts, not the magnitudes, are the point. First, the two scalar policies fail on *different* cells, the fixed source can never carry encoding, topical top-1 forfeits the trap, yet land within about one point of the same aggregate: the cap is the policy *class*, since any total order must abandon at least one cell, not the choice of scorer (Theorem 2, Proposition 2). Second, the policy that refuses to choose is the *worst* of the four: anti-monotonicity is the dominant cost (see Section 7.4.3). Third, the oracle shortfall from 100% is neither pure selection error nor pure capability. For DeepSeek (68–75% band) it is the model's own limit; for Haiku the shortfall on `api_post_ok` is not even that, since the retained completions (see Section 7.4.5) finds the deficit lying in an omitted `import` line, with W_1 scoring 40/40 once it is neutralized. The four cells were built to exhibit the phenomenon, so these magnitudes diagnose the failure structure rather than estimate production-traffic gains.

What follows for practice. Four statements survive the caveats above. (i) One number per source is structurally lossy: when two sources each carry a decisive fact, any single task-independent ranking drops one of them (Theorem 2). Keep both and route by task instead; the per-cell maximum in Table 5 is headroom read off the test outcomes, and no implementable router here captures it, on these tasks or new ones. Concatenating them is not forbidden by the theorem, but on this pair

Table 5: Mean PASS (%) over the four verified cells under four selection policies, per model, computed from the per-cell PASS counts behind Table A1 (averaging the rounded percentages instead would shift a column by up to 0.4). The last column is the per-cell maximum of W_1 and W_2 *read off the test outcomes*: an empirical oracle over the pair, not a router fixed in advance and then evaluated, and not a confidence bound on any policy's population quality. It is also not a ceiling over all selection policies, since it excludes W_{1+} as a candidate: admitting it raises the Haiku oracle from 82.5 to 95.0, because W_{1+} edges W_1 on the trap cell.

Model	Best fixed source	Topical top-1	Concatenate (W_{1+})	Oracle over $\{W_1, W_2\}$
Haiku-4.5	68.8	58.8	50.0	**82.5**
Sonnet-4.6	75.0	75.0	50.0	**100.0**
Opus-4.8	75.0	75.0	47.5	**100.0**
GPT-5.5	75.0	75.0	50.0	**100.0**
DeepSeek-V4-Pro	60.0	52.5	41.9	**76.9**
Kimi-K2.6	75.0	73.8	48.1	**98.8**
Mean	71.5	68.3	47.9	**93.0**

it is the worst of the four policies measured (Table 5). (ii) Do not cache a source's *value* across task families; cache retrieval representations instead, because the value is model- and task-relative (see Section 7.4.4). (iii) Adding correct, on-topic text to a working prompt can break it, by as much as 100% → 0%, so a regression after enlarging the context is a candidate anti-monotonicity collapse before it is a model failure (see Section 7.4.3). (iv) A cheap topical ranker fails exactly where the surface lexicon points at the wrong source, which is where a listwise reader earns its cost (Proposition 2). Each is measurable on any stack with the released harness.

8 Discussion

The results recast a piece of practitioner folklore, "relevance does not transfer across tasks", as a structural fact: when two sources are Blackwell-incomparable the order over them is partial, and any per-source scalar score, by imposing one task-independent preorder, must fail to rank them faithfully on some task. What follows concerns what that evidence does and does not license, and where the formalization is a model of the realized object rather than a proof about it.

8.1 Scope of the claims

Three limits of scope govern everything above. **(i) Two registers.** The theorems are about the ideal user of a source; every number we measure is the pass rate of one fixed, non-optimal rule δ_M. The two registers are established separately, not one from the other. Structural incomparability follows from the projection construction and Lemma 1, with no prior, no caps and no measurement. The experiments separately verify realized cross-advantage for the fixed model on the named task pairs (Assumption 1), which is $(M, \mathcal{D})$-incomparability and nothing more. Lemma 2 is a conditional Bayes-value criterion whose guessing-cap hypotheses these state-conditional measurements do not estimate. So the measured crossover does not *earn* the structural conclusion, and need not: for these sources Lemma 1 already supplies it by construction. **(ii) The estimator's name.** $\widehat{\delta}_{\mathcal{D}}$ does no minimization over garblings and ranges over a fixed set of tasks; it is a one-sided *value-deficiency surrogate*, not a Le Cam deficiency, and the two are not interchangeable for a fixed rule (Remark 8).

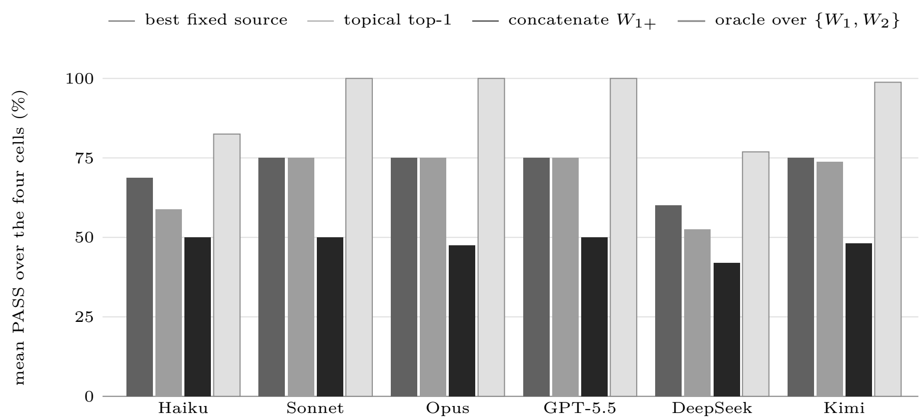

Figure 3: Mean PASS over the four verified cells under the four selection policies of Table 5, per model. Two things are visible that the table states in prose. The two scalar policies land within about a point of each other on most models while failing on *different* cells, so the ceiling is the policy class rather than the choice of scorer. And concatenating both sources is the *worst* of the four on every model, below even a fixed single source: refusing to choose costs more than choosing wrongly. The last column reads the per-cell maximum off the test outcomes and is an oracle over the pair, not a router fixed in advance and then evaluated.

What is rigorous is the verification, which we regard as the most robust part and which stands whether or not the deficiency analogy holds. **(iii) What the verification covers.** It is correct because it pays for picking the worst task with exact intervals and a union bound; its support is small (two of the four cells carry each model's result, and which two is model-specific: an API cell and the encoding cell on five models, the trap cell rather than an API cell on DeepSeek-V4-Pro), each family is controlled simultaneously at level η over the intervals it names, while the conjunction *across* families carries no single joint guarantee (see Section 6.5), and we call $\mathcal{D}$ a fixed adversarial battery, not a task distribution.

8.2 Anti-monotonicity is real, verified, but not capability-ordered

The harm object is the direct contrast $\Delta_T = \mathrm{PASS}(W_{1+}, T) - \mathrm{PASS}(W_1, T)$, which measures whether appending the (irrelevant-to-this-task) W_2 to a sufficient W_1 degrades performance (the second difference Ψ_T is an interaction term, not a harm measure; Remark 9). We verify $\Delta_T \leq -0.30$ on at least one cell for all six models (Table 4), with collapses as stark as $100\% \to 0\%$ (upper bound -0.85) on Sonnet, GPT-5.5 and Kimi. This is a verified empirical falsification of superset-*monotonicity* for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone scalar context-value score can represent.

Three cautions attach to it. The harm is robust across the capability ladder but not ordered by it (see Section 7.4.3), so we claim no capability trend. Which cell carries the verification is still a choice: we fix it per model before unblinding rather than picking the starkest afterwards, and cells differ in their W_1 ceiling, so magnitudes are not comparable across models even though each binary verdict stands on its own. And the two Opus rows join arms from different collection windows (see Section 7.3); the harm contrasts do not, since each is read within a single run.

8.3 Mechanism: a correlate, not an explanation

On every collapse cell for which we have token telemetry, the harmful arm inflates output between +93% and +520%; the re-measured Opus superset arm carries no token record, so it is outside this comparison (see Appendix C). On the still-passing encoding cell the same superset arm ranges from −73% to +114% with no consistent direction, so the association does not establish an internal cause and we treat it as a reproducible correlate of the harmful arm rather than a mechanism. Per-model figures are in Appendix C.

8.4 Limitations and threats to validity

We collect the scope restrictions here; each is stated once and sized to its weight.

From diagnosis to selection. The constructive selection rule the partial order implies, keep the verified non-dominated set and cover what it leaves uncovered, is left for future work; here we note only its cost. Building the verified order on m candidates costs $m{\cdot}|\mathcal{D}|{\cdot}n$ verifier-gated calls, curation scale, not corpus scale; scaling verified selection to retrieval over 10^6 passages, by amortizing the probe into a learned deficiency predictor or running the partial-order step as a re-ranking head over a scalar shortlist, is the central open problem this work opens. Keeping a non-dominated set instead of a top-k prefix already exists as an operator, the skyline retrieval step of Zhu et al. [7], and learned set selectors optimize evidence sets end to end [26, 27]; what the partial order adds is the object the set is taken over, verified task value rather than heuristic per-item scores. We do not claim that retaining the non-dominated set makes downstream selection *safe* under anti-monotonicity, and nothing here proves it: the verification bounds the relation *between* the candidates, not what a selector then does with them, and concatenating two retained members is the move our own record shows can lower PASS (see Section 7.4.3). What the partial order contributes at this level is diagnostic.

Source-pair generality. Incomparability and dominance hold for two hand-built pairs in two domains, both measured over the same transport as the six-model record. On the second pair the verification returns $L_{\mathcal{D}} = +0.728/+0.586$, so the crossover is present in a `cache` module as well as a `ledger` one. The superset harms there too, though mildly: on the trap cell it takes V_1 from 72% to 65%, far short of the $\tau = 0.30$ margin, so what the second domain establishes is the crossover, not a second verified collapse. An earlier command-line measurement of this pair behaved differently enough that we reported it as regime-dependent; that caveat concerned a transport the record no longer uses. The cross-vendor record is still single-pair: five of the six models were never run on it. The superset harms that pair only mildly, well short of the τ margin, and the pair did not reproduce the relevance-trap effect at all, its prompt not being lexically V_2-leaning. A third domain and a vendor-general strong mis-rank remain open next steps.

Cross-vendor magnitudes are still only suggestive. All six models are now run the same way, one HTTP request per draw at $n = 40$ (see Section 7.3), so neither the turn budget nor the transport confounds the comparison. What remains is the ordinary caution that different vendors' models differ in ways we do not control (training data, size, tokenizer), so the *magnitude* comparisons (the value spectrum, the Δ_T ladder) should be read as suggestive; the binary within-model claims do not depend on any cross-vendor comparison.

The construction is an idealization. It models sources as clean projections onto independent coordinates; real chunks overlap, contradict, and leak (the measured θ-dependence is exactly such leakage), so it is the cleanest *sufficient* condition for incomparability, realized exactly by our hand-built pairs and not claimed of arbitrary ones.

Prevalence is not measured, and is the decisive next experiment. Every claim here is existential: it establishes *that* verified incomparability and superset harm occur, on which models, and with which confounds excluded. It does not establish *how often* either occurs in a deployed retrieval stream, and no number in this paper should be read as an estimate of that. The four cells were built to make the structure visible, so their aggregate magnitudes are diagnostic of the failure mode rather than of production traffic. The direct experiment is specified: draw document pairs from a natural corpus, keep the procedure of Section 6 but write tasks and verifiers that corpus admits, since the four ledger and wire tests do not transfer, and report what fraction verifies incomparable and harmful, with the sampling frame stated. That fraction is a lower bound on prevalence at the stated n and thresholds, not prevalence itself. We name it as the open question rather than approximate it here, because a prevalence figure computed from hand-built pairs would be circular.

Scope of the behavioural controls. The padding, routing-note and structured-context arms, and the reversed-order arm, were each run over the same API path as the main record, on Haiku at $n = 40$ on the two collapse cells, in the same regime and on the same task versions as the main table. None of the four is part of the six-model record, so what these controls establish is established on one model, not across the roster. The padding match is by *characters* (1108 against 1095 at the arm level), not by subword tokens, and because W_2 is symbol-dense while the filler is plain prose the two arms are not tokenizer-matched; token count is therefore *not* isolated, only gross length. The filler is also off-topic for the `ledger` module. Taken together, on this model the added block's content and the routing instruction move the outcome, while equal character-length off-topic filler does not. The controls constrain positional and gross-length explanations in the tested configuration rather than eliminating them: they do not isolate tokenization or processing difficulty, do not identify a unique cause, and do not exclude every position-dependent mechanism, since reversing the two blocks moves the `api_post_ok` superset from 10/40 to 0/40, so order still bears on the size of the observed harm. The mechanism and refusal counts of Appendix C, with the non-empty extraction record, are likewise automated flags over retained draws: they document what the failing replies contain, establish no universal causal mechanism, and do not exclude every packaging problem. A tokenizer-matched control and an on-topic but verifier-irrelevant control remain untested, as does any of these across the roster. The order control has been run, but on one model, over the command-line interface, and on the two collapse cells only (see Section 7.4.3).

Sampling and verification regime. The verification assumes n i.i.d. Bernoulli PASS draws per cell (Assumption 2). Each draw is a separate call with no shared session or conversation history (protocol in Appendix D), which makes independence plausible; it does not *establish* it, nor that the served backend was stationary across a run. Both remain assumptions on which the coverage statements depend, and a provider-side model or routing change during a run would break them silently. Stationarity *between* runs rests on less still, and the cross-window contrasts of Section 7.3 depend on it; the within-run harm contrasts do not. Marginal-versus-joint control is discussed in Section 8.1.

Ranker spectrum and synergy. The trap mis-rank spans lexical cosine and three representative open-weight learned rankers, with an LLM listwise reranker ranking the pair correctly, which the theory permits rather than predicts, but the learned checkpoints are the small members of their families and the evidence is one trap in one domain. The mis-rank is not evidence of class membership: we do not verify that any learned checkpoint's score is invariant to the verifier-decisive coordinate or increasing in lexical overlap alone, so Proposition 2 is not shown to bind these rankers. Positive interference (synergy) is untested by construction: W_1, W_2 are task-disjoint, so a complementary pair would be needed to test whether a superset can also *help* beyond its parts; we make no claim about it.

9 Conclusion

Context selection is a partial-order problem, not a scalar one. When two sources are incomparable, no single per-source score ranks them correctly for all tasks (Theorem 2), and a value-deficiency surrogate with finite-sample bounds makes that order measurable on a live model. On six models across four vendors we verify at $\geq 90\%$ that two private sources cross in realized pass rate; on each of them a deployed relevance score ranks the pair the wrong way on a designed trap; and on all twelve cells adding a strictly larger source lowers the pass rate, as far as $100\% \to 0\%$. What a private source is worth is model-relative: guessability runs from 0% to 75% across the six models, and what it tracks these runs cannot identify. Bringing Blackwell and Le Cam to inference-time context is not ours to claim; concurrent work reached the same apparatus from the admission side (see Section 2.3). What follows from taking *incomparability* rather than deficiency-minimization as the object is the impossibility, and the instrument that tests it.

The scope is narrow and stated exactly. The theorems concern the idealized Bayes-optimal user; the measured crossover and collapse are properties of a *fixed* model; the two sources' incomparability is structural, resting on the projection construction and Lemma 1 rather than on measurement; Lemma 2 is a conditional Bayes-value criterion whose guessing caps we do not estimate; and the estimator is a regret-gap surrogate, not a Le Cam deficiency. All six models are measured under one protocol, so cross-vendor magnitudes carry no transport confound; they remain read with caution as model differences and collection windows (see Section 7.3); the verified claims are within-model contrasts. What the impossibility excludes is one number per source fixed across tasks and query-conditioned scores meeting the two hypotheses of Proposition 2; a task-conditioned score, a router over public metadata, or a listwise reader that pays to read the decisive content lie outside it (Remark 6). The decisive next test is prevalence: draw source pairs from a natural corpus rather than building them, run the same verification unchanged, and report how often a deployed retrieval stream produces a pair that verifies incomparable (see Section 8.4).

Author Contributions

Conceptualization, A.B. and V.L.; methodology, A.B., V.L. and P.P.; software, A.B.; validation, A.B., V.L. and Z.R.; formal analysis, A.B. and P.P.; investigation, A.B.; resources, V.L.; data curation, A.B.; writing—original draft preparation, A.B.; writing—review and editing, A.B., V.L., P.P. and Z.R.; visualization, A.B. and Z.R.; supervision, V.L. and P.P.; project administration, V.L. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement

Not applicable. The study involved no human participants and no animals; all measurements are automated queries to commercial language-model endpoints, graded by executable verifiers.

Informed Consent Statement

Not applicable.

Data Availability Statement

All data and code supporting the reported results are openly available at https://github.com/abilous-ti/blackwell-llm-context. The repository contains the measurement harness (Python standard library only), every per-model run log and result file cited in Appendix B, and the raw-completion diagnostic records of Section 7.4.5. Two kinds of record must be distinguished. The original runs stored *aggregated* per-cell counts and logs; raw model completions were not retained, and cannot be recovered. The retained completions of Appendix C.1 and the diagnostic runs stored *raw* completions, and those are released in full. Claims that depend on reading replies are therefore made only against the second kind. `results/MANIFEST.md` pins every released record by SHA-256 and maps each table and figure to the files it is computed from; it certifies the integrity of what is published, not the existence of what was never kept. The repository's full commit history is published with it, so the provenance of every artifact is inspectable. No data were withheld; every reported figure is computed from those files by the released scripts.

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Anthropic's Claude models were used for code scaffolding, experimental-plan review, implementation assistance, consistency and artifact audits, manuscript organization, and LaTeX formatting. The authors inspected code changes, executed the experiments, verified outputs against frozen manifests and source files, checked cited sources, and take full responsibility for the study design, results, interpretations, and manuscript.

Use of Generative Artificial Intelligence

Language models appear in this work in two distinct roles, and we separate them because only one bears on the results. *As the object of study:* the measured quantities are pass rates of commercial language models, obtained through the interfaces described in Section 7.3; everything else we report, the confidence bounds, the ranker scores, the token and cost figures, is computed from those runs. *As a tool:* the assistance described in the Acknowledgments. No result, table or figure was produced by a language model acting as an author or an analyst: every reported quantity is computed by the released scripts from the released records, which `results/MANIFEST.md` pins by SHA-256 and maps to the table or figure that uses it. The Data Availability Statement says which records are aggregate-only and which retain raw completions.

Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BSS	Blackwell–Sherman–Stein (theorem)
CP	Clopper–Pearson (exact binomial confidence interval)
DPP	Determinantal point process
EIG	Expected information gain
IR	Information retrieval
LLM	Large language model
MMR	Maximal marginal relevance
PASS	Pass rate of a fixed model rule on a verifier-graded task
PRP	Probability ranking principle
RAG	Retrieval-augmented generation

A Deferred proofs and technical material

Proof of Lemma 1

Proof. (a) For deterministic experiments a garbling G with $\kappa_{W_B} = G \circ \kappa_{W_A}$ must compute $\pi_B(S)$ from $\pi_A(S)$ alone (no further access to S). Such a measurable map exists iff π_B is $\sigma(\pi_A)$-measurable, i.e. $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$. Under coordinate independence and non-degeneracy, $\sigma(\pi_B) \subseteq \sigma(\pi_A)$ holds iff every coordinate in B is among those in A, i.e. $B \subseteq A$: if some $j \in B \setminus A$, then S_j is independent of $\pi_A(S)$ and non-constant, so it is not $\sigma(\pi_A)$-measurable, and no garbling can reconstruct it; conversely if $B \subseteq A$ the coordinate-deletion map is a (deterministic) garbling. *(b)* is the contrapositive of (a) applied in both directions. $\square$

Proof of Lemma 2

Proof. (a) Suppose, for contradiction, $W_2 \succeq_B W_1$. By Theorem 1, $V^\star(W_2, T) \geq V^\star(W_1, T)$ for every task T, in particular at T_a. By the sufficiency hypothesis, T_a's verifier is determined by its decisive coordinate, which lies in $A_1 \setminus A_2$ and is revealed exactly by W_1 (Definition 3). Solvability, the hypothesis's separate half, supplies μ_{T_a}-almost surely an action the verifier accepts, measurable in that coordinate and the facts known μ_{T_a}-almost surely. Reading the coordinate off W_1's signal and playing that action attains $v_{T_a} = 1$ μ_{T_a}-almost surely, so $V^\star(W_1, T_a) = 1$. By hypothesis $g_a = V^\star(W_2, T_a) < 1$, hence $V^\star(W_1, T_a) > V^\star(W_2, T_a)$, a contradiction. Therefore $W_2 \not\succeq_B W_1$; symmetrically $g_b < 1$ forbids $W_1 \succeq_B W_2$, so W_1, W_2 are Blackwell-incomparable.

Two things this argument does *not* use are worth stating, because the natural shorter proof uses both and is wrong. It never asserts $V^\star(W_1, T_a) \geq \mathrm{PASS}_M(W_1, T_a)$: the left side is a Bayes value under μ_{T_a}, the right a pass rate at the single realized convention, and optimality relates $V^\star$ only to the μ_{T_a}-averaged $\overline{\mathrm{PASS}}_M$ (Remark 1). And it never treats the measured crossover as the hypothesis or as an estimate of it: the caps are assumed, and nothing reported here determines g_a or g_b. In the converse direction the caps keep apart two failures a bare crossover cannot distinguish: opposed PASS_M rates on a strictly *comparable* pair, which is what (W_1, W_{1+}) gives, with $W_{1+} \succeq_B W_1$ strictly and Theorem 3's non-coherence a monotonicity violation; and opposed rates under Blackwell *equivalence*, illustrated without a certified instance by the padding arm (W_1, W_{pad}) (Remarks 3 and 2). Neither licenses a structural conclusion from a crossover alone.

(b) Assumption 1 directly yields $\mathrm{PASS}_M(W_1, T_a) > \mathrm{PASS}_M(W_2, T_a)$ and the reverse at T_b, which is Definition 5's incomparability. The unconditional converse fails, witnessed by the decoder that ignores its context: whatever the sources, $\mathrm{PASS}_M(W_1, T) = \mathrm{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$, so $W_1 \succeq_{M,\mathcal{D}} W_2$ and $W_2 \succeq_{M,\mathcal{D}} W_1$ both hold and the pair is $(M, \mathcal{D})$-comparable however Blackwell-incomparable it is. A monotonicity violation on a *comparable* pair is a different phenomenon, not by itself such a witness. $\square$

Proof of Theorem 2

Proof. By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, "not $W_1 \succeq_B W_2$" yields a task T_b (a prior plus bounded utility) with $V^\star(W_2, T_b) > V^\star(W_1, T_b)$, and "not $W_2 \succeq_B W_1$" yields a task T_a with $V^\star(W_1, T_a) > V^\star(W_2, T_a)$; both inequalities are *strict.* These witnesses carry bounded utility and need not be binary verifiers (Definition 2); for projection sources they may be taken binary (proof of Lemma 2(a)). A query-independent φ assigns the single ordered pair $(\varphi(W_1), \varphi(W_2))$, fixing one task-independent total *preorder* on $\{W_1, W_2\}$. If $\varphi(W_1) = \varphi(W_2)$ it expresses no strict preference and is unfaithful on both; otherwise the strict order it fixes agrees with the optimal-value order on at most one of T_a, T_b (they point opposite ways) and is violated on the other. Either way faithfulness fails on at least one, which is the claim, and a fixed task-independent tie-breaking rule only removes the first case. The PASS_M conclusion follows by replacing $V^\star$ with PASS_M via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin γ. □

Proof of Proposition 2

Proof. For (a), $\varphi = g \circ \psi$ and $\psi(W_1, T_c) = \psi(W_2, T_c)$ force equal scores, while faithfulness on T_c would require a strict gap. For (b), $\mathrm{ov}(W_2, T_c) > \mathrm{ov}(W_1, T_c)$ with g strictly increasing gives $\varphi(W_2, T_c) > \varphi(W_1, T_c)$, while utility favours W_1 because T_c's PASS requires W_1's coordinate; the two orders disagree. Part (b) makes explicit the two things the informal reading of "saturated lexicon" presumes, strict monotonicity and overlap being the only feature read, without which no strict inequality follows. □

Proof of Proposition 1

Proof. Let G be the garbling with $\kappa_{W_2} = G \circ \kappa_{W_1}$. Any predictor $g \in \mathcal{V}$ acting on W_2's signal is realized on W_1's signal by the composite $g \circ G$, which lies in $\mathcal{V}$ by the closure hypothesis. Therefore the best $\mathcal{V}$-predictor under W_1 attains at least the predictive accuracy of the best under W_2, so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under W_1. The scalarity is immediate: $I_\mathcal{V}$ returns a single real number per source, so it induces one task-independent total preorder and, by Theorem 2, cannot rank a Blackwell-incomparable pair faithfully on all tasks. □

Proof of Proposition 3

Proof. Let E be the event that all $2|\mathcal{D}|$ Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most $\alpha = \eta/(2|\mathcal{D}|)$ (exact coverage of Clopper–Pearson [9]), so by the union bound $\Pr[E^c] \le 2|\mathcal{D}| \cdot \alpha = \eta$, giving $\Pr[E] \ge 1 - \eta$. On E, for every T we have $p_{W_2,T} \ge L_{W_2,T}$ and $p_{W_1,T} \le U_{W_1,T}$, so $p_{W_2,T} - p_{W_1,T} \ge L_{W_2,T} - U_{W_1,T}$; taking $\max_T$ on both sides, $\sup_T [p_{W_2,T} - p_{W_1,T}] \ge L_\mathcal{D}(W_1, W_2)$. Thus on E (probability $\ge 1 - \eta$) the true sup is at least $L_\mathcal{D}(W_1, W_2)$, which is the claimed lower confidence bound; if it is positive, the true sup is positive, verifying $d_{M,\mathcal{D}}(W_1, W_2) > 0$. The reverse direction uses $L_{W_1,T}$ and $U_{W_2,T}$, which are endpoints of the *same* $2|\mathcal{D}|$ intervals already included in E; therefore both one-sided verifications are valid on the single event E, and their conjunction holds at $\ge 1 - \eta$ without a further Bonferroni factor. □

Proof of Proposition 4

Proof. Only two quantities enter $\overline{\Delta}_{T^\star}$: an upper endpoint on $p_{W_{1+},T^\star}$ and a lower endpoint on $p_{W_1,T^\star}$. The two arms are independent Bernoulli samples (separate completions). Let E_+ be the event $p_{W_{1+},T^\star} \le U_{W_{1+},T^\star}$ and E_- the event $p_{W_1,T^\star} \ge L_{W_1,T^\star}$. By the exact one-sided coverage of Clopper–Pearson at noncoverage $\alpha = \eta/2$ per endpoint, $\Pr[E_+^c] \le \eta/2$ and $\Pr[E_-^c] \le \eta/2$, so by the union bound $\Pr[E_+ \cap E_-] \ge 1 - \eta$. On that event

$$\Delta_{T^\star} \;=\; p_{W_{1+},T^\star} - p_{W_1,T^\star} \;\le\; U_{W_{1+},T^\star} - L_{W_1,T^\star} \;=\; \overline{\Delta}_{T^\star},$$

which is the claim; if $\overline{\Delta}_{T^\star} \le -\tau$ then $\Delta_{T^\star} \le -\tau$ at confidence $\ge 1 - \eta$. Note that the budget is spent on exactly the two endpoints used: a two-sided interval at level $\eta/2$ would place $\eta/4$ in each tail, leave two endpoints unused, and deliver $1 - \eta/2$ while widening the bound, which is why the allocation is stated per endpoint rather than per interval. For a post-hoc selected cell, replacing α by $\eta/(2|\mathcal{D}|)$ and union-bounding over the $2|\mathcal{D}|$ endpoints preserves $\geq 1-\eta$ simultaneous coverage. $\square$

Proof of Theorem 3

Proof. $W_{1+} \succeq_B W_1$ by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)); write G for that deletion, so $G \circ \kappa_{W_{1+}}(\cdot \mid s) = \kappa_{W_1}(\cdot \mid s)$ for every state s. The rule $\delta_M \circ G$ is then Blackwell-coherent on $T^\star$ in the sense of the theorem, and its realized pass rate on W_{1+} equals $\mathrm{PASS}_M(W_1, T^\star)$ at the realized state, since the two rules see the same signal law. The hypothesis $\mathrm{PASS}_M(W_{1+}, T^\star) < \mathrm{PASS}_M(W_1, T^\star)$ therefore says that δ_M on W_{1+} is strictly beaten by its own pruned version on the same input, so δ_M is not Blackwell-coherent on $T^\star$. Note what this argument does *not* use: it is the simulation construction behind (i)⇒(ii) of Theorem 1 applied state-conditionally, not an instance of statement (ii), which is Bayes-averaged and so says nothing about a single state (Remark 1). A monotone scalar ϕ assigns $\phi(W_{1+}) \geq \phi(W_1)$ (monotonicity), hence orders W_{1+} above W_1 and cannot reproduce the realized reversal. $\square$

Relation to LLM-evaluation statistics

Remark 10 (Relation to LLM-evaluation statistics)*.* The verification sits in a small but growing statistical-methodology literature for black-box model evaluation: paired-difference designs and clustered standard errors for evals [70], two-sample tests verifying behavioral differences between API-served models [71], and the recommendation to prefer exact frequentist intervals over CLT approximations when per-cell n is below a few hundred [72], which is precisely our regime ($n \in [20, 80]$ per cell) and precisely what the exact Clopper–Pearson construction provides.

Power and the role of n

Both verifications are driven by how close the arms sit to an endpoint. For a stark cell with empirical proportion at an endpoint ($\hat{p} \in \{0, 1\}$) the Clopper–Pearson interval becomes one-sided and clears 0 even at small n. It does not become a point: at $n = 40$ and two-sided level $\eta/8$, 0/40 gives [0, 0.119] and 40/40 gives [0.881, 1], and it is that 0.119 the crossing has to beat. The incomparability verification (6) is carried by exactly such cells (W_1 at $\approx 100\%$, W_2 at 0% and vice versa). The harm contrast (8) needs the gap between two arms to exceed τ after both intervals are paid for, so a mild collapse from a non-endpoint arm can be verified in sign without clearing $\tau = 0.30$ at $n = 40$: the second source pair, whose superset takes the trap cell from 72% to 65%, is such a case.

B Extended experimental record

Generality and controls: full protocols

Four follow-up experiments address three threat axes: source-pair generality, the context-length confound, and whether the anti-monotonicity collapse is curable by cheap prompting.

A second incomparable pair, different domain. We repeated the protocol with a second hand-built pair on a private `cache` module: V_1, the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False "newly inserted" flag, a `CacheError` type), versus V_2, a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at $n = 40$ the pair is again verified incomparable ($L_{\mathcal{D}} = +0.728/+0.586$, both directions clear of zero) with a clean double crossover (V_1 solves the contract and trap cells, V_2 solves

Table A1: Per-cell PASS rates (%) for all four verified cells, by arm, all one HTTP turn at $n = 40$. Every model was measured on every cell and arm. The encoding rows (`enc_amount`) carry the $W_2 > W_1$ direction of incomparability: on every model W_2 solves encoding while W_1 fails it, mirroring the API cells where W_1 solves and W_2 fails.

Model	Cell	`none`	W_1	W_2	W_{1+}
Haiku-4.5	`api_post_ok`	0	100	0	0
Haiku-4.5	`api_argorder`	0	80	0	0
Haiku-4.5	`enc_amount`	0	0	55	100
Haiku-4.5	`trap_store_wire`	0	95	0	100
Sonnet-4.6	`api_post_ok`	0	100	0	0
Sonnet-4.6	`api_argorder`	0	100	0	0
Sonnet-4.6	`enc_amount`	0	0	100	100
Sonnet-4.6	`trap_store_wire`	0	100	0	100
Opus-4.8	`api_post_ok`	0	100	0	0
Opus-4.8	`api_argorder`	100	100	0	0
Opus-4.8	`enc_amount`	0	0	100	90
Opus-4.8	`trap_store_wire`	0	100	0	100
GPT-5.5	`api_post_ok`	5	100	0	0
GPT-5.5	`api_argorder`	75	100	0	0
GPT-5.5	`enc_amount`	55	0	100	100
GPT-5.5	`trap_store_wire`	0	100	0	100
DeepSeek-V4-Pro	`api_post_ok`	0	75	0	0
DeepSeek-V4-Pro	`api_argorder`	2	68	0	5
DeepSeek-V4-Pro	`enc_amount`	0	0	68	65
DeepSeek-V4-Pro	`trap_store_wire`	0	98	0	98
Kimi-K2.6	`api_post_ok`	0	100	0	0
Kimi-K2.6	`api_argorder`	45	100	0	0
Kimi-K2.6	`enc_amount`	0	0	95	92
Kimi-K2.6	`trap_store_wire`	0	100	0	100

key-normalization, all none-baselines floor at 0%). Anti-monotonicity reproduces in direction only: the superset takes the trap cell from 72% to 65%, which does not approach the margin does not clear the margin $\tau = 0.30$ fixed for the whole programme. The binding constraint is the V_1 ceiling itself (80%, versus pair-1's $\sim$100%), so the margin is met with little to spare; at $n = 40$ the same collapse gives -0.25 and does not clear, which is why the cell was re-run at $n = 80$. The $n = 80$ re-run covers the `cache_put_ok` cell alone, so its dominance control ($U_{\mathcal{D}} = 0.000$ with a verified strict win) is a statement about that cell rather than about the three-cell battery; the battery-wide control remains the $n = 40$ one. The relevance trap for this pair was a null, its prompt vocabulary leaned toward V_1 ($\varphi(V_1) = 0.39 > \varphi(V_2) = 0.25$), so relevance *agreed* with PASS, so the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability and dominance results to a new domain*, not a second mis-rank.

A character-matched padding control. Anti-monotonicity compares W_1 against the strictly longer W_{1+}, and added input length alone is known to degrade performance [43], so the collapse could in principle be a token-count effect rather than a signal effect. To isolate this we added a padding arm W_{pad}: filler text in the same "team context (private)" register as W_2, placed in the same position W_2 occupies in W_{1+} (padding first, W_1 last), character-matched to W_{1+} within 13 characters, but carrying no coordinate any verifier checks (office-logistics content). On Haiku at $n = 40$ on the two collapse cells, padding is free while the equal-length dominated signal is catastrophic: on `api_argorder`, W_1 98%, W_{pad} 100%, W_{1+} 10%; on `api_post_ok`, W_1 50%, W_{pad} 65%, W_{1+} 10%; and the same-run interference verification clears on `api_argorder` ($\Delta_T = -0.88$, upper bound -0.67). In the tested configuration the collapse therefore does not follow from added length alone. The match is by characters, not subword tokens, and the filler is plain prose while W_2 is symbol-dense, so gross length is constrained but token count and processing difficulty are not. One residual axis is not isolated by this control: the padding is off-topic for the `ledger` module while W_2 is on-topic, so "topical but verifier-irrelevant" text remains a smaller, untested middle case.

A routing-instruction ablation: the cure presupposes the diagnosis. If the collapse reflects convention confusion, an explicit routing note might prevent it, and if a one-line prompt fully cured the harm, its practical significance would shrink. We measured this directly: the arm W_{1+}^{instr} appends to W_{1+} a single routing sentence stating which convention serves which task family ("for tasks that call the ledger API use only the API call contract; the wire-encoding invariant applies only to encode/decode tasks"). On Haiku at $n = 40$ the instruction recovers most, but not all, of the collapse: on `api_argorder`, 2% $\to$ 80% (W_1 alone: 100%), and on `api_post_ok`, 22% $\to$ 68% (W_1 alone: 50%); both recoveries over W_{1+} are large (non-overlapping exact 95% intervals), while on the carrying cell the instructed superset still sits below W_1 alone (80% vs. 100%). That residual gap is verified negative by the same two-arm contrast used throughout, $\Delta_T = -0.20$ with upper bound -0.03, but it falls far short of the harm margin $\tau = 0.30$: the instruction removes the collapse without closing the gap. We pre-commit to the two readings jointly. First, the harm is substantially mediated by convention interference that explicit routing knowledge prevents, so pipelines that can attach per-task routing labels have a cheap mitigation. Second, and decisively for the paper's thesis: *writing the routing sentence requires already knowing which source is decisive for which task family*, which is precisely the coordinate structure the partial order formalizes and the deficiency verification measures; the cure is an injection of selection-time coordinate knowledge into the prompt, not a scalar score, and no query-independent scalar can supply it (Theorem 2). Formally, δ_M with the routing note is a *different, more Blackwell-coherent rule*; the verified falsification of Theorem 3 concerns the base rule and is unaffected, and even under the instructed rule, keeping the dominated signal is at best equal and directionally worse than dropping it, so the selection-level conclusion, keep the non-dominated set rather than forcing the superset, survives the ablation intact.

A structured-context ablation: labeled XML segmentation mitigates only weakly. To separate "the model cannot segment the two conventions" from "the model cannot route them to task families," the arm W_{1+}^{xml} wraps the same two conventions, in the same order as W_{1+}, in labeled XML sections (<wire_encoding_invariant>, <api_call_contract>) with no routing statement. On Haiku at $n = 40$ per cell, structure alone does not mitigate at all: `api_argorder` 0% (vs. W_{1+} 0%, W_{1+}^{instr} 80%, W_1 100%) and `api_post_ok` 25% (vs. 12%, 68%, 57%). The structure-only mitigation is separated from the raw superset on the carrying cell (exact 95% intervals $[0.293, 0.615]$ vs. $[0.001, 0.132]$) but remains far below W_1 alone; on `api_post_ok` it is not separated from the raw superset. Protocol note: the W_{1+}^{xml} arm was measured in a clean arm-only re-run (its arm in the first run was invalidated by a transport-failure window and scored no completions; the re-run had zero transport errors), so its comparison against same-day baseline arms is across runs rather than within one run. The resulting mitigation hierarchy, raw superset $<$ structure $<$ routing $<$

Table A2: Retained completions on the two collapse cells, $n = 40$ per arm. "Import" counts replies carrying the `import ledger` line the task needs; "empty" counts replies from which no code block could be extracted; "refusal" counts replies declining the task. Mean length is in characters of the raw reply.

Model	Cell	Arm	Mean length	Import	Empty / Refusal
Haiku-4.5	`api_post_ok`	W_1	105	40/40	0 / 0
		W_{1+}	477	39/40	0 / 0
	`api_argorder`	W_1	466	33/40	0 / 0
		W_{1+}	642	40/40	0 / 0
Sonnet-4.6	`api_post_ok`	W_1	93	40/40	0 / 0
		W_{1+}	381	40/40	0 / 0
	`api_argorder`	W_1	178	40/40	0 / 0
		W_{1+}	251	40/40	0 / 0
Opus-4.8	`api_post_ok`	W_1	92	40/40	0 / 0
		W_{1+}	285	40/40	0 / 0
	`api_argorder`	W_1	102	40/40	0 / 0
		W_{1+}	250	40/40	0 / 0

dropping the dominated signal, is consistent with the operative variable being convention-to-task routing rather than segmentation alone; as in Section 8.3 this localization is an interpretation of the behavioural contrasts, not a verified claim, and it rests on one model at $n = 40$ with the structure arm scored in a separate run.

C Experimental diagnostics

This appendix carries the per-draw diagnostic record behind Section 7.4.5 and the token observation of Section 8.3. Nothing here is required to read the verified results; it is the evidence for what the failing replies contain.

C.1 What the retained completions contain

Every draw behind the interference table was kept. This appendix reads them. The table below covers the two collapse cells of the three Anthropic models, 40 completions per arm and cell, 480 in total; the same retention is in place for the behavioural controls and the second source pair.

Three readings follow and one is withheld. *Packaging is excluded*: no draw produced an empty extraction, and the `import` line is present in 39 or 40 of 40 replies everywhere except Haiku's W_1 arm on `api_argorder` (33/40), which is the arm that passes, so its omissions cannot explain the superset's failures. *The harmful arm inflates output on every model*: 105 → 477 and 466 → 642 characters on Haiku, 93 → 381 and 178 → 251 on Sonnet, 92 → 285 and 102 → 250 on Opus. *No model declined*: refusal language appears in none of the 480 draws, so the refusal behaviour an earlier command-line record showed for Sonnet is not present here.

What is withheld is the mechanism. A strict test for the superset reply applying W_2's wire encoding to a plain-cents amount fires on 0/80 and 1/80 of Haiku's failing superset replies, so this record excludes the packaging explanations without establishing what replaces them. Theorem 3 does not depend on the answer: it constrains any rule whose PASS falls under a strict superset, whatever the internal cause.

C.2 Output-token inflation on the harmful arm

Every collapse is accompanied by output-token inflation on the harmful arm. On the two API collapse cells W_{1+} emits between +93% and +520% more output tokens than W_1 (Haiku $+115\%/+163\%$; Sonnet $+520\%/+473\%$; GPT-5.5 $+281\%/+274\%$; DeepSeek $+93\%/+166\%$; Kimi $+203\%/+249\%$; the re-measured Opus W_{1+} arm carries no token record). One might read this as "verbose thrashing" causing the failure, but the relationship is correlational and uncontrolled: on the *still-passing* encoding cell the same superset arm ranges from -73% to $+114\%$ across the same models, with no consistent direction, which undercuts a clean "verbosity causes collapse" account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

One qualification governs this whole subsection. The retained completions (Appendix C.1) exclude the packaging explanations but do not identify what replaces them, and no draw on any model declined the task. The mechanism discussion below therefore describes nine cells, not all eleven.

The claim structure is deliberately mechanism-agnostic: Theorem 3 verifies that the realized rule is not Blackwell-coherent *whatever* the internal cause, and the behavioral controls constrain the hypothesis space from outside, the collapse survives reversal of the block order (order control) and is not reproduced by an off-topic block of equal character length (padding control), and is largely curable by explicit convention routing (instruction ablation), a signature consistent with mis-binding rather than capacity exhaustion. That the one-line routing note recovers most of the collapse while labeled XML segmentation of the same content recovers only a minority ($2\% \to 45\%$ against $\to 80\%$) locates the operative variable in convention-to-task routing rather than formatting; this is an interpretation, not a finding, and every verified claim is independent of it. An activation-level account is out of reach here, since four of the six models are closed-weight and the phenomenon is cross-vendor; the direct route is to reproduce the collapse on an open-weight model with the same harness and instrument the contrasting arms, downstream of rather than prerequisite to the behavioral claim. Distractor interference in RAG has been localized to a sparse feature subspace and corrected there [47]; ours differs in cause, the added source is each true, and each needed by some task in the battery though not by every task rather than a distractor, and it is not a knowledge conflict in the sense of Xu et al. [48], since nothing in W_1 and W_2 contradicts.

D Reproducibility and protocol

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The complete artifact bundle (the measurement harness, the executable verifiers, and every per-model run log and result JSON cited below) is available at `https://github.com/abilous-ti/blackwell-llm-context`. The harness (`harness/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a stdlib regularized incomplete beta), is cross-OS, and implements the surrogate $\widehat{\delta}_{\mathcal{D}}$, the uniform incomparability verification (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference verification, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic Messages API, the Azure Responses API, and OpenAI-compatible chat completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

Protocol. The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type);

return code 0 is PASS. The four verified cells are `api_post_ok`, `api_argorder` (`memo` arrives via a parameter, forcing the keyword-only fact), `enc_amount`, and `trap_store_wire`; the arms are $\{\text{none}, W_1, W_2, W_{1+}\}$, with W_{pad} (character-matched neutral filler $+\!\!+\, W_1$) added for the padding control. All six models are queried over HTTP (one request, one turn, the prompt instructing tool abstention rather than a flag enforcing it; code extracted from the returned text), and PASS is measured over $n = 40$ i.i.d. completions per cell at $\eta = 0.10$.

Decoding, seeding, and retries. All runs use the provider's default decoding configuration: the harness sets no explicit temperature, top-p, or seed (the launchers are invoked without sampling overrides), so the n draws per cell are independent samples from each provider's default decoder, each a fresh single-turn call with a 200-second timeout. This describes every run reported in the paper. Earlier versions of this record reached the Anthropic models, the four behavioural controls and the second source pair through the command line; each has since been re-measured over the API path, and every number reported here comes from those runs, whose logs carry the `[HTTP-API]` marker. The one exception is the raw-failure audit of Section 7.4.5, which explains an anomaly of the superseded command-line record and therefore cannot be re-measured: it is what that record was.

Four arm-only re-measurements were originally produced by ad-hoc scripts that were not kept, so their regime could be asserted but not recovered from a run log. All four have since been re-measured with `harness/diag/rerun_arm.py`, which calls the same `run_one` the six-model tables are built from and writes a log recording the regime. Three reproduced their published values closely: the Opus W_2/`trap_store_wire` and Haiku W_1/`enc_amount` cells returned 0/40 exactly as before, and the structured-context arm moved by one draw on one cell. The Opus W_{1+} arm moved further, and deeper: 38% $\to$ 12% on `api_post_ok` and 15% $\to$ 5% on `api_argorder`. This manuscript reports the re-measurements throughout, so every run cited here has a log naming its regime; `harness/diag/check_regimes.py` lists them with the evidence for each—an inventory of the listed runs, not a proof that the list is complete or that each regime is disclosed where its numbers are used. The between-run movement on the Opus arm is itself a data point for Section 8.4: repeated measurement of the same cell months apart shifts by more than the nominal interval width. Transport-level failures (empty or zero-token results) are retried up to three times with quadratic backoff; a draw that still fails is scored as a failure, never as a pass. Where a transport-outage window hit a whole arm or a single cell, we re-measured it cleanly in isolation with the retry harness and report only the clean measurement: the Opus W_{1+} arm; the Opus W_2/`trap_store_wire` cell, roughly 18 of whose 40 first-run draws were outage-scored (inferred from per-draw cost); and the Haiku W_1/`enc_amount` cell, whose first run predated the retry wrapper. One cell resisted re-measurement for longer than the others: GPT-5.5 W_2/`trap_store_wire` returned HTTP 500 after retries during the original run and no GPT-5.5 deployment was available on the resource at first revision, so for one revision cycle its reported 0/40 was carried with outage-scored draws in it. A GPT-5.5 deployment was subsequently provisioned and all four GPT-5.5 cells were re-run single-shot at $n = 40$ per arm with retention and zero transport failures. The re-run reproduces the published grid exactly in all eight W_1/W_2 entries, the disputed cell included (W_2/`trap_store_wire` 0/40 clean), so every cell in the coverage family is now a valid binomial sample and the reported n is the number of real draws throughout.

Artifacts. Per-model run logs and result JSONs are in the artifact repository (`https://github.com/abilous-ti/blackwell-llm-context`) under the pattern `blackwell_*.{log,json}`: the six single-turn $n = 40$ model runs, each a complete file with no per-arm override (`blackwell_haiku_api_n40`, `blackwell_sonnet_api_n40`, `blackwell_opus_api_n40`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, `blackwell_kimi_n40`), and the follow-up controls. All four behavioural controls share one run, `blackwell_controls_api_n40` (640 calls), whose arms are the padding filler, the reversed concatenation, the routing note and the XML segmentation measured against the same W_1 and W_{1+} baselines; `blackwell_pair2_api_n40` carries the second source pair and every $L_{\mathcal{D}}$ and Δ_T reported for it; `reranker_trap_api_n100` is the LLM listwise reranker probe, which records each judgment so the halfway checkpoint needs no second run (`harness/measure_reranker_trap.py`); and `dense_trap.json` (dense/cross-encoder probe, deterministic local scoring of open-weight checkpoints at zero API cost; `harness/measure_dense_trap.py`). The full per-cell record is in `docs/EXPERIMENT-BLACKWELL.md` and the theory hardening (formalization and proof sketches) in `docs/BLACKWELL.md`.

Compute and cost. The reported record is roughly 8,000 model calls: 640 per model across the six-model grid, 640 for the behavioural controls, and the second pair and reranker probe on top. We do not report a spend figure: the HTTP endpoints used here return no cost field, so any total would be an estimate from token counts rather than a measurement. The dense/cross-encoder probe ran locally on open weights at zero API cost.

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